

## Class: Character

| Method       | # | Test Description                                           | Sample Input Data                                     | Expected Output                                     | Actual Output                                       | P/F |
|--------------|---|------------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------|-----|
| GenerateID() | 1 | Automatically assigns an ID upon instantiation             | Create a new Character object                         | Character's ID is not null or 0                     | Character's ID is not null or 0                     | P   |
|              | 2 | Formats the ID correctly based on the yyMMddHHmmss pattern | Create a new Character object                         | ID is exactly a 12-digit number                     | ID is exactly a 12-digit number                     | P   |
|              | 3 | Ensures uniqueness                                         | Create Character A, wait 1 second, create Character B | Character A's ID and Character B's ID are different | Character A's ID and Character B's ID are different | P   |

## Class: Pirate

| Method                   | # | Test Description                             | Sample Input Data                                              | Expected Output                                                           | Actual Output                                                                | P/F |
|--------------------------|---|----------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------|-----|
| AddBounty(int<br>Bounty) | 1 | Adds a positive amount to the current bounty | Pirate A (Current<br>Bounty: 50,000,000),<br>Input: 10,000,000 | Pirate A's bounty<br>becomes 60,000,000                                   | Pirate A's bounty<br>becomes<br>60,000,000                                   | P   |
|                          | 2 | Adds zero to the bounty                      | Pirate A (Current<br>Bounty: 50,000,000),<br>Input: 0          | Pirate A's bounty<br>remains 50,000,000                                   | Pirate A's bounty<br>remains<br>50,000,000                                   | P   |
|                          | 3 | Prevents adding a negative bounty amount     | Pirate A (Current<br>Bounty: 50,000,000),<br>Input: -5,000,000 | Prints "Cannot add a<br>negative bounty",<br>bounty remains<br>50,000,000 | Prints "Cannot<br>add a negative<br>bounty", bounty<br>remains<br>50,000,000 | P   |

## Class: PirateHunter

| Method                  | # | Test Description                             | Sample Input Data                      | Expected Output                         | Actual Output                           | P/F |
|-------------------------|---|----------------------------------------------|----------------------------------------|-----------------------------------------|-----------------------------------------|-----|
| AddCaptures(int Amount) | 1 | Adds a valid positive amount of captures     | PirateHunter A (0 Captures), Input: 5  | PirateHunter A's captures increase to 5 | PirateHunter A's captures increase to 5 | P   |
|                         | 2 | Adds zero captures to the total              | PirateHunter A (5 Captures), Input: 0  | PirateHunter A's captures remain 5      | PirateHunter A's captures remain 5      | P   |
|                         | 3 | Rejects adding a negative amount of captures | PirateHunter A (5 Captures), Input: -2 | Prints error message, captures remain 5 | Prints error message, captures remain 5 | P   |

## Class: DevilFruit

| Method         | # | Test Description                                             | Sample Input Data                 | Expected Output                                                  | Actual Output                                                    | P/F |
|----------------|---|--------------------------------------------------------------|-----------------------------------|------------------------------------------------------------------|------------------------------------------------------------------|-----|
| TriggerReinc() | 1 | Executes standard reincarnation upon owner death             | Devil Fruit with 1 Current Owner  | Current Owner becomes null, Character added to Historical Owners | Current Owner becomes null, Character added to Historical Owners | P   |
|                | 2 | Clears the Devil Fruit attribute from the deceased character | Devil Fruit with 1 Current Owner  | Deceased character's DFPower attribute becomes null              | Deceased character's DFPower attribute becomes null              | P   |
|                | 3 | Safely executes when the fruit has no current owner          | Devil Fruit with 0 Current Owners | Method safely completes without throwing an error                | Method safely completes without throwing an error                | P   |
| GenerateID()   | 1 | Automatically assigns an ID upon instantiation               | Create a new Devil Fruit object   | Character's ID is not null or 0                                  | Character's ID is not null or 0                                  | P   |

|                                                |   |                                                                   |                                                            |                                                                                 |                                                                                 |   |
|------------------------------------------------|---|-------------------------------------------------------------------|------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---|
|                                                | 2 | Formats the ID correctly based on the yyMMddHHmmss pattern        | Create a new Devil Fruit object                            | ID is exactly a 12-digit number                                                 | ID is exactly a 12-digit number                                                 | P |
|                                                | 3 | Ensures uniqueness                                                | Create Devil FruitA, wait 1 second, create Character B     | Character A's ID and Character B's ID are different                             | Character A's ID and Character B's ID are different                             | P |
| AddHistoricalOwner(Character HistoricalOwner ) | 1 | Successfully adds a former user to the historical owners list     | Fruit A (Empty History), Input: Character A                | Character A is added to the HistoricalOwners list                               | Character A is added to the HistoricalOwners list                               | P |
|                                                | 2 | Appends a second former user without overwriting previous history | Fruit A (History contains Character A), Input: Character B | HistoricalOwners list size becomes 2, contains both Character A and Character B | HistoricalOwners list size becomes 2, contains both Character A and Character B | P |
|                                                | 3 | Safely handles attempting to add a null character                 | Fruit A, Input: null                                       | List size remains the same, prints error or safely ignores                      | List size remains the same, prints error or safely ignores                      | P |

## Class: PirateCrew

| Method                           | # | Test Description                                   | Sample Input Data                                            | Expected Output                                                   | Actual Output                               | P/F |
|----------------------------------|---|----------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------------|-----|
| CalculateTotalCrewBounty()       | 1 | Determines total bounty with all active members    | 2 Pirates with 50,000,000 bounty each, both Status "Free"    | 100000000                                                         | 100000000                                   | P   |
|                                  | 2 | Determines total bounty excluding captured members | 1 Pirate (Free, 50,000,000), 1 Pirate (Captured, 50,000,000) | 50000000                                                          | 50000000                                    | P   |
|                                  | 3 | Determines total bounty of an empty crew           | 0 Pirate members                                             | 0                                                                 | 0                                           | P   |
| AddCrewMember(Pirate CrewMember) | 1 | Adds a pirate not in any crew                      | Pirate A (No Crew)                                           | Pirate A is added to CrewMembers list, Role becomes "Crew Member" | Pirate A is added to CrewMembers list, Role | P   |

|              |   |                                                            |                                |                                                                              |                                                                              |   |
|--------------|---|------------------------------------------------------------|--------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------|---|
|              |   |                                                            |                                |                                                                              | becomes "Crew Member"                                                        |   |
|              | 2 | Transfers a pirate from an old crew                        | Pirate A (In Crew 2)           | Pirate A is removed from Crew 2, added to Crew 1, Role becomes "Crew Member" | Pirate A is removed from Crew 2, added to Crew 1, Role becomes "Crew Member" | P |
|              | 3 | Prevents adding a pirate already in the crew               | Pirate A (Already in Crew 1)   | Prints "[Name] is already part of the [Crew Name]"                           | Prints "[Name] is already part of the [Crew Name]"                           | P |
| GenerateID() | 1 | Automatically assigns an ID upon instantiation             | Create a new PirateCrew object | Character's ID is not null or 0                                              | Character's ID is not null or 0                                              | P |
|              | 2 | Formats the ID correctly based on the yyMMddHHmmss pattern | Create a new PirateCrew object | ID is exactly a 12-digit number                                              | ID is exactly a 12-digit number                                              | P |

|                                    |   |                                                              |                                                                          |                                                                                                         |                                                                                                         |   |
|------------------------------------|---|--------------------------------------------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|---|
|                                    | 3 | Ensures uniqueness                                           | Create PirateCrew A, wait 1 second, create PirateCrew B                  | Character A's ID and Character B's ID are different                                                     | Character A's ID and Character B's ID are different                                                     | P |
| RemoveCrewMember(PirateCrewMember) | 1 | Successfully removes a standard crew member                  | Crew 1 with Pirate A (Role: "First Mate"), Input: Pirate A               | Pirate A is removed from CrewMembers list, Pirate A's Crew becomes null, Pirate A's role becomes "None" | Pirate A is removed from CrewMembers list, Pirate A's Crew becomes null, Pirate A's role becomes "None" | P |
|                                    | 2 | Successfully removes the Captain and unsets the role         | Crew 1 with Pirate B (Role: "Captain", IsCaptain: true), Input: Pirate B | Pirate B is removed, IsCaptain becomes false, Crew 1's Captain variable becomes null                    | Pirate B is removed, IsCaptain becomes false, Crew 1's Captain variable becomes null                    | P |
|                                    | 3 | Safely handles attempting to remove a pirate not in the crew | Crew 1, Input: Pirate C (Belongs to Crew 2)                              | List size remains the same, prints error or safely ignores                                              | List size remains the same, prints                                                                      | P |

|  |  |  |  |  |                         |  |
|--|--|--|--|--|-------------------------|--|
|  |  |  |  |  | error or safely ignores |  |
|--|--|--|--|--|-------------------------|--|

**Class: MarineCorp**

| Method                           | # | Test Description                         | Sample Input Data   | Expected Output                                                 | Actual Output                                                   | P/F |
|----------------------------------|---|------------------------------------------|---------------------|-----------------------------------------------------------------|-----------------------------------------------------------------|-----|
| AddCorpMember(Marine CorpMember) | 1 | Adds a Marine not currently in any corps | Marine A (No Corps) | Marine A is added to CorpMembers list, Marine A's corps updates | Marine A is added to CorpMembers list, Marine A's corps updates | P   |

|                                 |   |                                                |                                |                                                        |                                                        |   |
|---------------------------------|---|------------------------------------------------|--------------------------------|--------------------------------------------------------|--------------------------------------------------------|---|
|                                 | 2 | Transfers a Marine currently in another corps  | Marine A (In Corps 2)          | Marine A is removed from Corps 2, added to Corps 1     | Marine A is removed from Corps 2, added to Corps 1     | P |
|                                 | 3 | Prevents adding a Marine already in this corps | Marine A (Already in Corps 1)  | List size remains the same, Marine A is not duplicated | List size remains the same, Marine A is not duplicated | P |
| AddOperationalFunds(int Amount) | 1 | Adds a positive amount to funds                | Corps 1 (0 Funds), Input: 5000 | OperationalFunds becomes 5000                          | OperationalFunds becomes 5000                          | P |
|                                 | 2 | Adds zero to funds                             | Corps 1 (5000 Funds), Input: 0 | OperationalFunds remains 5000                          | OperationalFunds remains 5000                          | P |

|                                     |   |                                          |                                                                                         |                                                                                                                 |                                                                                                                          |   |
|-------------------------------------|---|------------------------------------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|---|
|                                     | 3 | Evaluates adding a negative amount       | Corps 1 (5000 Funds),<br>In put: -1000                                                  | Prints error message,<br>funds remain 5000                                                                      | Prints error<br>message, funds<br>remain 5000                                                                            | P |
| RemoveCorpMember(Marine CorpMember) | 1 | Successfully removes a standard Marine   | Corps 1 with Marine A,<br>Input: Marine A                                               | Marine A is removed<br>from CorpMembers<br>list, Marine A's<br>MCorps becomes null                              | Marine A is<br>removed from<br>CorpMembers<br>list, Marine A's<br>MCorps becomes<br>null                                 | P |
|                                     | 2 | Successfully removes the Corps Commander | Corps 1 with Marine B<br>(Currently assigned as<br>CorpsCommander ),<br>Input: Marine B | Marine B is removed,<br>Marine B's MCorps<br>becomes null, Corps<br>1's CorpsCommander<br>variable becomes null | Marine B is<br>removed, Marine<br>B's MCorps<br>becomes null,<br>Corps 1's<br>CorpsCommander<br>variable<br>becomes null | P |

|  |   |                                                               |                                               |                                                            |                                                            |   |
|--|---|---------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|------------------------------------------------------------|---|
|  | 3 | Safely handles attempting to remove a Marine not in the corps | Corps 1, Input: Marine C (Belongs to Corps 2) | List size remains the same, prints error or safely ignores | List size remains the same, prints error or safely ignores | P |
|--|---|---------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|------------------------------------------------------------|---|

### Class: CharacterDatabase

| Method                  | # | Test Description                       | Sample Input Data                                               | Expected Output                                                                                                                  | Actual Output                                                                                                                    | P/F |
|-------------------------|---|----------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-----|
| addCharacter(Character) | 1 | Adding a single valid Character object | addCharacter(newObj) with a properly constructed Character      | characterList size increases by 1 and contains that exact object reference                                                       | characterList size increases by 1 and contains that exact object reference                                                       | P   |
|                         | 2 | Adding the same Character object twice | addCharacter(sameObj) called twice with the identical reference | characterList accepts the duplicate (size increases by 2, both entries referencing the same object) - no uniqueness check exists | characterList accepts the duplicate (size increases by 2, both entries referencing the same object) - no uniqueness check exists | P   |

|                 |   |                                            |                                                                                            |                                                                                                                                                         |                                                                                                                                                         |   |
|-----------------|---|--------------------------------------------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|                 | 3 | Passing null as the argument               | addCharacter(null)                                                                         | characterList.add(null) succeeds without throwing, since ArrayList permits null elements - this method performs no null validation                      | characterList.add(null) succeeds without throwing, since ArrayList permits null elements - this method performs no null validation                      | P |
| writeDatabase() | 1 | characterList contains one or more entries | writeDatabase() called after adding valid objects                                          | A snapshot of characterList is written to characterdatabase.dat no exception is thrown                                                                  | A snapshot of characterList is written to characterdatabase.dat no exception is thrown                                                                  | P |
|                 | 2 | characterList is empty                     | writeDatabase() called with no entries ever added                                          | An empty list is still saved successfully to characterdatabase.dat                                                                                      | An empty list is still saved successfully to characterdatabase.dat                                                                                      | P |
|                 | 3 | The database file cannot be written to     | characterdatabase.dat is read-only, locked by another process, or its directory is missing | The resulting IOException is caught console prints "Error: " + the exception's message the method returns normally instead of propagating the exception | The resulting IOException is caught console prints "Error: " + the exception's message the method returns normally instead of propagating the exception | P |

|                |   |                                                                           |                                                                                                                 |                                                                                                                                                           |                                                                                                                                                           |   |
|----------------|---|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| readDatabase() | 1 | characterdatabase.dat contains a previously saved, valid list             | readDatabase() called after a prior successful writeDatabase()                                                  | characterList is replaced in memory with the loaded list read from characterdatabase.dat                                                                  | characterList is replaced in memory with the loaded list read from characterdatabase.dat                                                                  | P |
|                | 2 | characterdatabase.dat exists but is empty (zero-length, never written to) | readDatabase() called on a freshly created, never-written file                                                  | An EOFException is caught console prints "Error: " + the exception's message the in-memory list retains its previous value rather than being nulled out   | An EOFException is caught console prints "Error: " + the exception's message the in-memory list retains its previous value rather than being nulled out   | P |
|                | 3 | The file contains corrupted data or an incompatible saved class version   | characterdatabase.dat content is truncated, corrupted, or was written by an older/incompatible class definition | A StreamCorruptedException, InvalidClassException, or ClassNotFoundException is caught console prints "Error: " + the exception's message no crash occurs | A StreamCorruptedException, InvalidClassException, or ClassNotFoundException is caught console prints "Error: " + the exception's message no crash occurs | P |

### Class: DevilFruitDatabase

| Method            | # | Test Description                        | Sample Input Data                                        | Expected Output                                                                                                            | Actual Output                                                                                                              | P/F |
|-------------------|---|-----------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|-----|
| addDF(DevilFruit) | 1 | Adding a single valid DevilFruit object | addDF(newObj) with a properly constructed DevilFruit     | dftList size increases by 1 and contains that exact object reference                                                       | dftList size increases by 1 and contains that exact object reference                                                       | P   |
|                   | 2 | Adding the same DevilFruit object twice | addDF(sameObj) called twice with the identical reference | dftList accepts the duplicate (size increases by 2, both entries referencing the same object) - no uniqueness check exists | dftList accepts the duplicate (size increases by 2, both entries referencing the same object) - no uniqueness check exists | P   |

|                 |   |                                        |                                                                                             |                                                                                                                                                         |                                                                                                                                                         |   |
|-----------------|---|----------------------------------------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|                 | 3 | Passing null as the argument           | addDF(null)                                                                                 | dftList.add(null) succeeds without throwing, since ArrayList permits null elements - this method performs no null validation                            | dftList.add(null) succeeds without throwing, since ArrayList permits null elements - this method performs no null validation                            | P |
| writeDatabase() | 1 | dftList contains one or more entries   | writeDatabase() called after adding valid objects                                           | A saved snapshot of dftList is written to devilfruitdatabase.dat no exception is thrown                                                                 | A saved snapshot of dftList is written to devilfruitdatabase.d at no exception is thrown                                                                | P |
|                 | 2 | dftList is empty                       | writeDatabase() called with no entries ever added                                           | An empty list is still saved successfully to devilfruitdatabase.dat                                                                                     | An empty list is still saved successfully to devilfruitdatabase.d at                                                                                    | P |
|                 | 3 | The database file cannot be written to | devilfruitdatabase.dat is read-only, locked by another process, or its directory is missing | The resulting IOException is caught console prints "Error: " + the exception's message the method returns normally instead of propagating the exception | The resulting IOException is caught console prints "Error: " + the exception's message the method returns normally instead of propagating the exception | P |

|                |   |                                                                            |                                                                                                                  |                                                                                                                                                           |                                                                                                                                                           |   |
|----------------|---|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| readDatabase() | 1 | devilfruitdatabase.dat contains a previously saved, valid list             | readDatabase() called after a prior successful writeDatabase()                                                   | dftList is replaced in memory with the loaded list read from devilfruitdatabase.dat                                                                       | dftList is replaced in memory with the loaded list read from devilfruitdatabase.dat                                                                       | P |
|                | 2 | devilfruitdatabase.dat exists but is empty (zero-length, never written to) | readDatabase() called on a freshly created, never-written file                                                   | An EOFException is caught console prints "Error: " + the exception's message the in-memory list retains its previous value rather than being nulled out   | An EOFException is caught console prints "Error: " + the exception's message the in-memory list retains its previous value rather than being nulled out   | P |
|                | 3 | The file contains corrupted data or an incompatible saved class version    | devilfruitdatabase.dat content is truncated, corrupted, or was written by an older/incompatible class definition | A StreamCorruptedException, InvalidClassException, or ClassNotFoundException is caught console prints "Error: " + the exception's message no crash occurs | A StreamCorruptedException, InvalidClassException, or ClassNotFoundException is caught console prints "Error: " + the exception's message no crash occurs | P |

## Class: MarineCorpDatabase

| Method                | # | Test Description                        | Sample Input Data                                           | Expected Output                                                                                                                | Actual Output                                                                                                                  | P/F |
|-----------------------|---|-----------------------------------------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-----|
| addMCorp(Marine Corp) | 1 | Adding a single valid MarineCorp object | addMCorp(newObj) with a properly constructed MarineCorp     | MCorpList size increases by 1 and contains that exact object reference                                                         | MCorpList size increases by 1 and contains that exact object reference                                                         | P   |
|                       | 2 | Adding the same MarineCorp object twice | addMCorp(sameObj) called twice with the identical reference | MCorpList accepts the duplicate (size increases by 2, both entries referencing the same object) - no uniqueness check exists   | MCorpList accepts the duplicate (size increases by 2, both entries referencing the same object) - no uniqueness check exists   | P   |
|                       | 3 | Passing null as the argument            | addMCorp(null)                                              | MCorpList.add(null) succeeds without throwing, since ArrayList permits null elements - this method performs no null validation | MCorpList.add(null) succeeds without throwing, since ArrayList permits null elements - this method performs no null validation | P   |

|                 |   |                                                                |                                                                                             |                                                                                                                                                         |                                                                                                                                                         |   |
|-----------------|---|----------------------------------------------------------------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| writeDatabase() | 1 | MCorpList contains one or more entries                         | writeDatabase() called after adding valid objects                                           | A saved snapshot of MCorpList is written to marinecorpdatabase.dat no exception is thrown                                                               | A saved snapshot of MCorpList is written to marinecorpdatabase.dat no exception is thrown                                                               | P |
|                 | 2 | MCorpList is empty                                             | writeDatabase() called with no entries ever added                                           | An empty list is still saved successfully to marinecorpdatabase.dat                                                                                     | An empty list is still saved successfully to marinecorpdatabase.dat                                                                                     | P |
|                 | 3 | The database file cannot be written to                         | marinecorpdatabase.dat is read-only, locked by another process, or its directory is missing | The resulting IOException is caught console prints "Error: " + the exception's message the method returns normally instead of propagating the exception | The resulting IOException is caught console prints "Error: " + the exception's message the method returns normally instead of propagating the exception | P |
| readDatabase()  | 1 | marinecorpdatabase.dat contains a previously saved, valid list | readDatabase() called after a prior successful writeDatabase()                              | MCorpList is replaced in memory with the loaded list read from marinecorpdatabase.dat                                                                   | MCorpList is replaced in memory with the loaded list read from marinecorpdatabase.dat                                                                   | P |

|  |   |                                                                            |                                                                                                                  |                                                                                                                                                           |                                                                                                                                                           |   |
|--|---|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|  | 2 | marinecorpdatabase.dat exists but is empty (zero-length, never written to) | readDatabase() called on a freshly created, never-written file                                                   | An EOFException is caught console prints "Error: " + the exception's message the in-memory list retains its previous value rather than being nulled out   | An EOFException is caught console prints "Error: " + the exception's message the in-memory list retains its previous value rather than being nulled out   | P |
|  | 3 | The file contains corrupted data or an incompatible saved class version    | marinecorpdatabase.dat content is truncated, corrupted, or was written by an older/incompatible class definition | A StreamCorruptedException, InvalidClassException, or ClassNotFoundException is caught console prints "Error: " + the exception's message no crash occurs | A StreamCorruptedException, InvalidClassException, or ClassNotFoundException is caught console prints "Error: " + the exception's message no crash occurs | P |

### Class: PirateCrewDatabase

| Method                | # | Test Description                        | Sample Input Data                                           | Expected Output                                                                                                              | Actual Output                                                                                                                | P/F |
|-----------------------|---|-----------------------------------------|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|-----|
| addPCrew(Pirate Crew) | 1 | Adding a single valid PirateCrew object | addPCrew(newObj) with a properly constructed PirateCrew     | PCrewList size increases by 1 and contains that exact object reference                                                       | PCrewList size increases by 1 and contains that exact object reference                                                       | P   |
|                       | 2 | Adding the same PirateCrew object twice | addPCrew(sameObj) called twice with the identical reference | PCrewList accepts the duplicate (size increases by 2, both entries referencing the same object) - no uniqueness check exists | PCrewList accepts the duplicate (size increases by 2, both entries referencing the same object) - no uniqueness check exists | P   |

|                 |   |                                        |                                                                                             |                                                                                                                                                         |                                                                                                                                                         |   |
|-----------------|---|----------------------------------------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|                 | 3 | Passing null as the argument           | addPCrew(null)                                                                              | PCrewList.add(null) succeeds without throwing, since ArrayList permits null elements - this method performs no null validation                          | PCrewList.add(null) succeeds without throwing, since ArrayList permits null elements - this method performs no null validation                          | P |
| writeDatabase() | 1 | PCrewList contains one or more entries | writeDatabase() called after adding valid objects                                           | A saved snapshot of PCrewList is written to piratecrewdatabase.dat no exception is thrown                                                               | A saved snapshot of PCrewList is written to piratecrewdatabase.dat no exception is thrown                                                               | P |
|                 | 2 | PCrewList is empty                     | writeDatabase() called with no entries ever added                                           | An empty list is still saved successfully to piratecrewdatabase.dat                                                                                     | An empty list is still saved successfully to piratecrewdatabase.dat                                                                                     | P |
|                 | 3 | The database file cannot be written to | piratecrewdatabase.dat is read-only, locked by another process, or its directory is missing | The resulting IOException is caught console prints "Error: " + the exception's message the method returns normally instead of propagating the exception | The resulting IOException is caught console prints "Error: " + the exception's message the method returns normally instead of propagating the exception | P |

|                |   |                                                                            |                                                                                                                  |                                                                                                                                                           |                                                                                                                                                           |   |
|----------------|---|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| readDatabase() | 1 | piratecrewdatabase.dat contains a previously saved, valid list             | readDatabase() called after a prior successful writeDatabase()                                                   | PCrewList is replaced in memory with the loaded list read from piratecrewdatabase.dat                                                                     | PCrewList is replaced in memory with the loaded list read from piratecrewdatabase.dat                                                                     | P |
|                | 2 | piratecrewdatabase.dat exists but is empty (zero-length, never written to) | readDatabase() called on a freshly created, never-written file                                                   | An EOFException is caught console prints "Error: " + the exception's message the in-memory list retains its previous value rather than being nulled out   | An EOFException is caught console prints "Error: " + the exception's message the in-memory list retains its previous value rather than being nulled out   | P |
|                | 3 | The file contains corrupted data or an incompatible saved class version    | piratecrewdatabase.dat content is truncated, corrupted, or was written by an older/incompatible class definition | A StreamCorruptedException, InvalidClassException, or ClassNotFoundException is caught console prints "Error: " + the exception's message no crash occurs | A StreamCorruptedException, InvalidClassException, or ClassNotFoundException is caught console prints "Error: " + the exception's message no crash occurs | P |

## Class: BountyManager

| Method                                     | # | Test Description                                                      | Sample Input Data                                                                                | Expected Output                                                                                                                                     | Actual Output                                                                                                                   | P/F |
|--------------------------------------------|---|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-----|
| registerCapture(Character, Pirate, String) | 1 | The target Pirate's status is not "Dead" (ex: "Alive" or "Captured")  | registerCapture(captor, target, "captured") where target.GetStatus() = "Alive"                   | A new CaptureRecord(captor, target) is constructed and added to captureHistory size increases by 1                                                  | A new CaptureRecord(captor, target) is constructed and added to captureHistory size increases by 1                              | P   |
|                                            | 2 | The target's status matches "dead" only by different casing           | target.GetStatus() = "DEAD" or "dEaD"                                                            | equalsIgnoreCase("dead") still matches regardless of case, so an IllegalStateException is thrown just as with exact-case "Dead"                     | equalsIgnoreCase("dead") still matches regardless of case, so an IllegalStateException is thrown just as with exact-case "Dead" | P   |
|                                            | 3 | The targetState parameter is passed but never consulted by the method | registerCapture(captor, target, null) or any arbitrary string for targetState, with target alive | The capture is still registered successfully the value (or null) passed for targetState has no effect on behavior or on the resulting CaptureRecord | The capture is still registered successfully the value (or null) passed for targetState has no effect on behavior               | P   |

|                 |   |                                              |                                                                                         |                                                                                                                                                         |                                                                                                                                                         |   |
|-----------------|---|----------------------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|                 |   |                                              |                                                                                         |                                                                                                                                                         | or on the resulting CaptureRecord                                                                                                                       |   |
|                 | 4 | The target Pirate's status is exactly "Dead" | registerCapture(captor, target, "captured") where target.GetStatus() = "Dead"           | An IllegalStateException("Cannot capture a pirate that is already dead") is thrown captureHistory is left unmodified (size unchanged)                   | An IllegalStateException("Cannot capture a pirate that is already dead") is thrown captureHistory is left unmodified (size unchanged)                   | P |
| writeDatabase() | 1 | captureHistory contains one or more entries  | writeDatabase() called after adding valid objects                                       | A saved snapshot of captureHistory is written to bountydatabase.dat no exception is thrown                                                              | A saved snapshot of captureHistory is written to bountydatabase.dat no exception is thrown                                                              | P |
|                 | 2 | captureHistory is empty                      | writeDatabase() called with no entries ever added                                       | An empty list is still saved successfully to bountydatabase.dat                                                                                         | An empty list is still saved successfully to bountydatabase.dat                                                                                         | P |
|                 | 3 | The database file cannot be written to       | bountydatabase.dat is read-only, locked by another process, or its directory is missing | The resulting IOException is caught console prints "Error: " + the exception's message the method returns normally instead of propagating the exception | The resulting IOException is caught console prints "Error: " + the exception's message the method returns normally instead of propagating the exception | P |

|                |   |                                                                        |                                                                |                                                                                                                                                         |                                                                                                                                                         |   |
|----------------|---|------------------------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| readDatabase() | 1 | bountydatabase.dat contains a previously saved, valid list             | readDatabase() called after a prior successful writeDatabase() | captureHistory is replaced in memory with the loaded list read from bountydatabase.dat                                                                  | captureHistory is replaced in memory with the loaded list read from bountydatabase.dat                                                                  | P |
|                | 2 | bountydatabase.dat exists but is empty (zero-length, never written to) | readDatabase() called on a freshly created, never-written file | An EOFException is caught console prints "Error: " + the exception's message the in-memory list retains its previous value rather than being nulled out | An EOFException is caught console prints "Error: " + the exception's message the in-memory list retains its previous value rather than being nulled out | P |

**Class: CaptureRecord**

| Method                  | # | Test Description                                   | Sample Input Data                                                                                 | Expected Output                                                                                                                                     | Actual Output                                                                                                                                       | P/F |
|-------------------------|---|----------------------------------------------------|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| routeFinancialRewards() | 1 | The captor is a Marine assigned to a MarineCorp    | nanghuli = a Marine whose GetMCorps() returns a valid MarineCorp<br>nahuli.GetBounty() = 50000000 | marineCaptor.GetMCorps().AddOperationalFunds(50000000) is called<br>console prints the amount added and the corps' base location                    | marineCaptor.GetMCorps().AddOperationalFunds(50000000) is called<br>console prints the amount added and the corps' base location                    | P   |
|                         | 2 | The captor is a Marine with no assigned MarineCorp | nanghuli = a Marine whose GetMCorps() returns null                                                | No funds are added anywhere<br>(AddOperationalFunds() is never called) console prints "Marine has no Corps to receive the funds."                   | No funds are added anywhere<br>(AddOperationalFunds() is never called) console prints "Marine has no Corps to receive the funds."                   | P   |
|                         | 3 | The captured Pirate's bounty is 0                  | nahuli.GetBounty() = 0,<br>nanghuli = a PirateHunter                                              | SetWallet() is still called with the captor's unchanged wallet value (+0) console still prints the (zero-amount) reward message no exception occurs | SetWallet() is still called with the captor's unchanged wallet value (+0) console still prints the (zero-amount) reward message no exception occurs | P   |
|                         | 4 | The captor is a PirateHunter or a Civilian         | nanghuli = a PirateHunter or Civilian,<br>nahuli.GetBounty() = 25000000                           | this.nanghuli.SetWallet(oldWallet + 25000000) is called console prints the amount added and the captor's name                                       | this.nanghuli.SetWallet(oldWallet + 25000000) is called console prints the amount added and the captor's name                                       | P   |

|                           |   |                                                                     |                                                                                           |                                                                                                                               |                                                                                                                               |   |
|---------------------------|---|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---|
| validateCaptor(Character) | 1 | The candidate captor is a Marine, Civilian, or PirateHunter         | validateCaptor(aMarine)<br>, validateCaptor(aCivilian)<br>, validateCaptor(aPirateHunter) | Returns true in all three cases                                                                                               | Returns true in all three cases                                                                                               | P |
|                           | 2 | The candidate captor is a Pirate                                    | validateCaptor(aPirate)                                                                   | Returns false, since Pirate is not one of the three authorized instanceof checks                                              | Returns false, since Pirate is not one of the three authorized instanceof checks                                              | P |
|                           | 3 | The candidate captor argument is null                               | validateCaptor(null)                                                                      | Each instanceof check safely evaluates to false for a null operand (no NullPointerException in Java) the method returns false | Each instanceof check safely evaluates to false for a null operand (no NullPointerException in Java) the method returns false | P |
| updateCapturedStatus()    | 1 | Called after a valid Pirate has been assigned to the captured field | this.nahuli = a valid Pirate<br>updateCapturedStatus() called                             | nahuli.SetStatus("captured") is invoked, updating that Pirate's status                                                        | nahuli.SetStatus("captured") is invoked, updating that Pirate's status                                                        | P |

|  |   |                                                        |                                                                   |                                                                                                            |                                                                                                            |   |
|--|---|--------------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|---|
|  | 2 | Called repeatedly for the same already-captured Pirate | updateCapturedStatus() invoked twice in a row for the same nahuli | setStatus("captured") is simply called again with the same value the operation is idempotent with no error | setStatus("captured") is simply called again with the same value the operation is idempotent with no error | P |
|--|---|--------------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|---|

### Class: CharacterController

| Method                                      | # | Test Description                                                                     | Sample Input Data                  | Expected Output                                                                                           | Actual Output                                                                                             | P/F |
|---------------------------------------------|---|--------------------------------------------------------------------------------------|------------------------------------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-----|
| actionPerformed(ActionEvent) - createButton | 1 | Clicking Create button sets info text and switches to the create panel               | characterView.createButton clicked | mainview.setInfoText("Creating a Character") is called<br>mainview.showPanel("CHARACTERCREATE") is called | mainview.setInfoText("Creating a Character") is called<br>mainview.showPanel("CHARACTERCREATE") is called | P   |
|                                             | 2 | Clicking Create button refreshes the Devil Fruit combo box before the panel is shown | characterView.createButton clicked | characterCreateController.refreshDFComboBox() is called prior to showPanel()                              | characterCreateController.refreshDFComboBox() is called prior to showPanel()                              | P   |

|                                                    |   |                                                                                  |                                        |                                                                                                    |                                                                                                       |   |
|----------------------------------------------------|---|----------------------------------------------------------------------------------|----------------------------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|---|
|                                                    | 3 | Clicking Create button refreshes crew/corp combo boxes before the panel is shown | characterView.createBut<br>ton clicked | characterCreateControll<br>er.refreshAffiliationCom<br>boBoxes() is called prior<br>to showPanel() | characterCreateCon<br>troller.refreshAffiliati<br>onComboBoxes() is<br>called prior to<br>showPanel() | P |
| actionPerformed(<br>ActionEvent) -<br>viewButton   | 1 | Clicking View button sets info text and switches to the view panel               | characterView.viewButto<br>n clicked   | setInfoText("Viewing a<br>Character") called<br>showPanel("CHARACT<br>ERVIEW") called              | setInfoText("Viewin<br>g a Character")<br>called<br>showPanel("CHAR<br>ACTERVIEW")<br>called          | P |
|                                                    | 2 | Clicking View button refreshes the character selection combo box                 | characterView.viewButto<br>n clicked   | characterViewController.<br>refreshComboBox() is<br>called before the panel<br>is shown            | characterViewContr<br>oller.refreshCombo<br>Box() is called<br>before the panel is<br>shown           | P |
| actionPerformed(<br>ActionEvent) -<br>modifyButton | 1 | Clicking Modify button sets info text and switches to the modify panel           | characterView.modifyBu<br>tton clicked | setInfoText("Modifying a<br>Character") called<br>showPanel("CHARACT<br>ERMODIFY") called          | setInfoText("Modifi<br>ng a Character")<br>called<br>showPanel("CHAR<br>ACTERMODIFY")<br>called       | P |

|                                             |   |                                                                        |                                    |                                                                                    |                                                                                    |   |
|---------------------------------------------|---|------------------------------------------------------------------------|------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|---|
|                                             | 2 | Clicking Modify button refreshes the character selection combo box     | characterView.modifyButton clicked | characterModifyController.refreshComboBox() is called before the panel is shown    | characterModifyController.refreshComboBox() is called before the panel is shown    | P |
|                                             | 3 | Clicking Modify button refreshes the Devil Fruit combo box             | characterView.modifyButton clicked | characterModifyController.refreshDFComboBox() is called before the panel is shown  | characterModifyController.refreshDFComboBox() is called before the panel is shown  | P |
| actionPerformed(ActionEvent) - deleteButton | 1 | Clicking Delete button sets info text and switches to the delete panel | characterView.deleteButton clicked | setInfoText("Unaliving a Character") called<br>showPanel("CHARACTERDELETE") called | setInfoText("Unaliving a Character") called<br>showPanel("CHARACTERDELETE") called | P |
|                                             | 2 | Clicking Delete button refreshes the character selection combo box     | characterView.deleteButton clicked | characterDeleteController.refreshComboBox() is called before the panel is shown    | characterDeleteController.refreshComboBox() is called before the panel is shown    | P |

|                                           |   |                                                                           |                                                               |                                                                                                                  |                                                                                                                  |   |
|-------------------------------------------|---|---------------------------------------------------------------------------|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|---|
| actionPerformed(ActionEvent) - backButton | 1 | Clicking Back button hides the character panel and resets info text       | characterView.backButton clicked                              | setInfoText("Main Menu") called<br>characterView.panel.setVisible(false) no<br>mainview.showPanel() call is made | setInfoText("Main Menu") called<br>characterView.panel.setVisible(false) no<br>mainview.showPanel() call is made | P |
|                                           | 2 | An ActionEvent from a source not registered to this controller is ignored | actionPerformed() called with an unrelated ActionEvent source | No info text change, no panel switch, and no exception is thrown                                                 | No info text change, no panel switch, and no exception is thrown                                                 | P |

### Class: CharacterCreateController

| Method                                    | # | Test Description                                                           | Sample Input Data                      | Expected Output                                                                       | Actual Output                                                                         | P/F |
|-------------------------------------------|---|----------------------------------------------------------------------------|----------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-----|
| actionPerformed(ActionEvent) - exitButton | 1 | Clicking Exit discards in-progress input and returns to the character menu | characterCreateView.exitButton clicked | characterCreateView.panel.setVisible(false)<br>mainview.showPanel("CHARACTER") called | characterCreateView.panel.setVisible(false)<br>mainview.showPanel("CHARACTER") called | P   |

|                                                                |   |                                                                               |                                                 |                                                                                                   |                                                                                                   |   |
|----------------------------------------------------------------|---|-------------------------------------------------------------------------------|-------------------------------------------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|---|
| actionPerformed(<br>ActionEvent) -<br>faction radio<br>buttons | 1 | Selecting Pirate reveals the Pirate sub-panel and the capture option          | characterCreateView.pirateRadioButton clicked   | captureRadioButton.setVisible(true)<br>showPanel("PIRATE") called                                 | captureRadioButton.setVisible(true)<br>showPanel("PIRATE") called                                 | P |
|                                                                | 2 | Selecting Marine defaults status to Alive and hides the capture option        | characterCreateView.marineRadioButton clicked   | aliveRadioButton selected<br>captureRadioButton.setVisible(false)<br>showPanel("MARINE") called   | aliveRadioButton selected<br>captureRadioButton.setVisible(false)<br>showPanel("MARINE") called   | P |
|                                                                | 3 | Selecting Pirate Hunter defaults status to Alive and hides the capture option | characterCreateView.hunterRadioButton clicked   | aliveRadioButton selected<br>captureRadioButton.setVisible(false)<br>showPanel("HUNTER") called   | aliveRadioButton selected<br>captureRadioButton.setVisible(false)<br>showPanel("HUNTER") called   | P |
|                                                                | 4 | Selecting Civilian defaults status to Alive and hides the capture option      | characterCreateView.civilianRadioButton clicked | aliveRadioButton selected<br>captureRadioButton.setVisible(false)<br>showPanel("CIVILIAN") called | aliveRadioButton selected<br>captureRadioButton.setVisible(false)<br>showPanel("CIVILIAN") called | P |

|                                             |   |                                                                       |                                                                     |                                                                                                                                     |                                                                                                                                     |   |
|---------------------------------------------|---|-----------------------------------------------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|---|
| actionPerformed(ActionEvent) - submitButton | 1 | Submitting a fully valid Pirate creates and persists a new character  | Name/Origin filled, pirateRadioButton selected, valid Bounty/Wallet | New Pirate object added via charDatabase.addCharacter() success JOptionPane shown panel hidden and mainview returns to "CHARACTER"  | New Pirate object added via charDatabase.addCharacter() success JOptionPane shown panel hidden and mainview returns to "CHARACTER"  | P |
|                                             | 2 | Submitting with a blank Name or blank Origin field is rejected        | nameTextField = "" (or originTextField = ""), other fields valid    | EmptyInputException caught error dialog reads "Name cannot be empty" or "Origin cannot be empty" respectively no character is added | EmptyInputException caught error dialog reads "Name cannot be empty" or "Origin cannot be empty" respectively no character is added | P |
|                                             | 3 | Submitting with non-numeric text in the Wallet field is rejected      | walletTextField = "abc"                                             | NumberFormatException caught dialog reads "Please ensure Wallet, Bounty, and Captures contain only numbers." no character is added  | NumberFormatException caught dialog reads "Please ensure Wallet, Bounty, and Captures contain only numbers." no character is added  | P |
|                                             | 4 | Selecting a real Devil Fruit assigns it and updates the fruit's owner | devilfruitComboBox index > 0                                        | newCharacter's dfPower is set to the chosen fruit dfPower.setCurrentOwner(newCharacter) is called                                   | newCharacter's dfPower is set to the chosen fruit dfPower.setCurrentOwner(newCharacter) is called                                   | P |

|  |   |                                                                       |                                                                |                                                                                                                                   |                                                                                                                                   |   |
|--|---|-----------------------------------------------------------------------|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|---|
|  | 5 | Leaving the Devil Fruit combo box on "None" assigns no fruit          | devilfruitComboBox index = 0                                   | newCharacter is created with dfPower = null<br>setCurrentOwner() is never called                                                  | newCharacter is created with dfPower = null<br>setCurrentOwner() is never called                                                  | P |
|  | 6 | Creating a Pirate with a crew selected adds them to that crew         | pirateRadioButton selected,<br>crewComboBox index > 0          | ((Pirate) newCharacter).SetPirateCrew(selectedCrew) is called with the matching PirateCrew object                                 | ((Pirate) newCharacter).SetPirateCrew(selectedCrew) is called with the matching PirateCrew object                                 | P |
|  | 7 | Creating a Marine with a corp selected assigns them to that corp      | marineRadioButton selected,<br>corpComboBox index > 0          | ((Marine) newCharacter).SetMCorps(selectedCorp) is called with the matching MarineCorp object                                     | ((Marine) newCharacter).SetMCorps(selectedCorp) is called with the matching MarineCorp object                                     | P |
|  | 8 | All four databases are saved on submit, whether it succeeds or throws | Submit once with valid data, once with walletTextField = "abc" | finally block calls writeDatabase() on charDatabase, devilFruitDatabase, pirateCrewDatabase, and marineCorpDatabase in both cases | finally block calls writeDatabase() on charDatabase, devilFruitDatabase, pirateCrewDatabase, and marineCorpDatabase in both cases | P |

|                                |   |                                                                             |                                                                                                   |                                                                                                       |                                                                                                       |   |
|--------------------------------|---|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|---|
| refreshDFComboBox()            | 1 | Refreshing the Devil Fruit dropdown repopulates it from the database        | devilFruitDatabase contains N fruits<br>refreshDFComboBox() called                                | devilfruitComboBox is cleared, item 0 is "None", followed by one entry per fruit as "ID - FruitName"  | devilfruitComboBox is cleared, item 0 is "None", followed by one entry per fruit as "ID - FruitName"  | P |
| refreshAffiliationComboBoxes() | 1 | Refreshing affiliation dropdowns repopulates both crew and corp combo boxes | pirateCrewDatabase and marineCorpDatabase each non-empty<br>refreshAffiliationComboBoxes() called | crewComboBox and corpComboBox are each cleared, item 0 is "None", followed by one entry per crew/corp | crewComboBox and corpComboBox are each cleared, item 0 is "None", followed by one entry per crew/corp | P |

### Class: CharacterViewController

| Method                                    | # | Test Description                                                     | Sample Input Data                     | Expected Output                                                                          | Actual Output                                                                            | P/F |
|-------------------------------------------|---|----------------------------------------------------------------------|---------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-----|
| actionPerformed(ActionEvent) - backButton | 1 | Clicking Back hides the view panel and returns to the character menu | characterViewView.back Button clicked | characterViewView.setFrame().setVisible(false)<br>mainview.showPanel("CHARACTER") called | characterViewView.setFrame().setVisible(false)<br>mainview.showPanel("CHARACTER") called | P   |

|                                                   |   |                                                                    |                                                                            |                                                                                                                 |                                                                                                                 |   |
|---------------------------------------------------|---|--------------------------------------------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|---|
| actionPerformed(ActionEvent) - charSelectComboBox | 1 | Selecting a valid character updates all shared attribute labels    | charSelectComboBox selection = valid index                                 | nameLabel, aliasLabel, originLabel, statusLabel, and walletLabel are updated with the selected character's data | nameLabel, aliasLabel, originLabel, statusLabel, and walletLabel are updated with the selected character's data | P |
|                                                   | 2 | The Devil Fruit label reflects whether the character owns a fruit  | One character with GetDFPower() == null, another with GetDFPower() != null | dfLabel reads "Devil Fruit: None" for the first and "Devil Fruit: " + fruit name for the second                 | dfLabel reads "Devil Fruit: None" for the first and "Devil Fruit: " + fruit name for the second                 | P |
|                                                   | 3 | Selecting a combo box index outside the database bounds is ignored | charSelectComboBox selection = -1 (or ≥ charDatabase size)                 | No labels are updated and no exception is thrown                                                                | No labels are updated and no exception is thrown                                                                | P |
|                                                   | 4 | Selecting a Pirate populates faction and bounty labels             | selectedChar instanceof Pirate                                             | specifiedLabel1 = "Faction: Pirate"<br>specifiedLabel2 = "Bounty: " + p.GetBounty()                             | specifiedLabel1 = "Faction: Pirate"<br>specifiedLabel2 = "Bounty: " + p.GetBounty()                             | P |

|  |   |                                                                                 |                                                        |                                                |                                                |   |
|--|---|---------------------------------------------------------------------------------|--------------------------------------------------------|------------------------------------------------|------------------------------------------------|---|
|  | 5 | Selecting a Pirate captain shows "Role: Captain" regardless of stored role text | selectedChar instanceof Pirate, p.IsCaptain() == true  | specifiedLabel3 = "Role: Captain"              | specifiedLabel3 = "Role: Captain"              | P |
|  | 6 | Selecting a non-captain Pirate shows their stored pirate role                   | selectedChar instanceof Pirate, p.IsCaptain() == false | specifiedLabel3 = "Role: " + p.GetPirateRole() | specifiedLabel3 = "Role: " + p.GetPirateRole() | P |
|  | 7 | Selecting a crewless Pirate shows "Crew: None"                                  | p.GetPirateCrew() == null                              | specifiedLabel4 = "Crew: None"                 | specifiedLabel4 = "Crew: None"                 | P |
|  | 8 | Selecting a Pirate with a crew shows the crew's name                            | p.GetPirateCrew() != null                              | specifiedLabel4 = "Crew: " + crew name         | specifiedLabel4 = "Crew: " + crew name         | P |

|  |    |                                                                                     |                                         |                                                                                                                             |                                                                                                                             |   |
|--|----|-------------------------------------------------------------------------------------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---|
|  | 9  | Selecting a Marine populates faction, rank, corp-commander flag, and corps          | selectedChar instance of Marine         | specifiedLabel1-4 show "Faction: Marine", rank, IsCorpCommander flag, and "Corps: None" or corp base location               | specifiedLabel1-4 show "Faction: Marine", rank, IsCorpCommander flag, and "Corps: None" or corp base location               | P |
|  | 10 | Selecting a Pirate Hunter populates faction, combat style, and captures             | selectedChar instance of PirateHunter   | specifiedLabel1-3 show "Faction: Pirate Hunter", combat style, and confirmed capture count<br>specifiedLabel4 remains blank | specifiedLabel1-3 show "Faction: Pirate Hunter", combat style, and confirmed capture count<br>specifiedLabel4 remains blank | P |
|  | 11 | Selecting a Civilian populates faction, profession, and residence                   | selectedChar instance of Civilian       | specifiedLabel1-3 show "Faction: Civilian", profession, and residence                                                       | specifiedLabel1-3 show "Faction: Civilian", profession, and residence                                                       | P |
|  | 12 | Switching from a Pirate to a Civilian clears the previous faction's leftover labels | Select a Pirate, then select a Civilian | specifiedLabel1-4 are all cleared before Civilian's 3 labels are set, leaving specifiedLabel4 blank                         | specifiedLabel1-4 are all cleared before Civilian's 3 labels are set, leaving specifiedLabel4 blank                         | P |

|                       |   |                                                                                              |                                                                 |                                                                                                                                               |                                                                                                                                               |   |
|-----------------------|---|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|---|
| refreshComboBox()<br> | 1 | Refreshing with characters present repopulates the dropdown and auto-selects the first entry | charDatabase contains ≥ 1 character<br>refreshComboBox() called | charSelectComboBox is cleared and repopulated as "ID - Name" a simulated ACTION_PERFORMED event fires, updating the labels for the first item | charSelectComboBox is cleared and repopulated as "ID - Name" a simulated ACTION_PERFORMED event fires, updating the labels for the first item | P |
|                       | 2 | Refreshing an empty database clears the dropdown without simulating a selection              | charDatabase is empty<br>refreshComboBox() called               | charSelectComboBox ends up empty no simulated ActionEvent is fired and no exception is thrown                                                 | charSelectComboBox ends up empty no simulated ActionEvent is fired and no exception is thrown                                                 | P |

### Class: CharacterModifyController

| Method                                    | # | Test Description                                                 | Sample Input Data                      | Expected Output                                                                       | Actual Output                                                                         | P/F |
|-------------------------------------------|---|------------------------------------------------------------------|----------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-----|
| actionPerformed(ActionEvent) - exitButton | 1 | Clicking Exit discards changes and returns to the character menu | characterModifyView.exitButton clicked | characterModifyView.panel.setVisible(false)<br>mainview.showPanel("CHARACTER") called | characterModifyView.panel.setVisible(false)<br>mainview.showPanel("CHARACTER") called | P   |

|                                                              |   |                                                                                             |                                                                                                    |                                                                                                                                                                                       |                                                                                                                                                                                                  |   |
|--------------------------------------------------------------|---|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| actionPerformed(<br>ActionEvent) -<br>charSelectCombo<br>Box | 1 | Selecting a character pre-fills the<br>universal fields and matching status<br>radio button | charSelectComboBox<br>selection = valid index,<br>GetStatus() is "Alive",<br>"Captured", or "Dead" | nameTextField,<br>aliasTextField,<br>originTextField, and<br>walletTextField show<br>current values the<br>matching status radio<br>button is selected                                | nameTextField,<br>aliasTextField,<br>originTextField, and<br>walletTextField<br>show current values<br>the matching status<br>radio button is<br>selected                                        | P |
|                                                              | 2 | Selecting a Pirate populates bounty,<br>role list, and pre-selects their current<br>crew    | selectedChar instanceof<br>Pirate                                                                  | specified1TextField<br>shows bounty<br>specified3ComboBox is<br>filled from Pirate.ROLES<br>specified4ComboBox<br>pre-selects the pirate's<br>current crew (or stays<br>on "None")    | specified1TextField<br>shows bounty<br>specified3ComboBo<br>x is filled from<br>Pirate.ROLES<br>specified4ComboBo<br>x pre-selects the<br>pirate's current crew<br>(or stays on "None")          | P |
|                                                              | 3 | Selecting a Marine populates the rank<br>list and pre-selects current rank and<br>corp      | selectedChar instanceof<br>Marine                                                                  | specified3ComboBox is<br>filled from<br>Marine.RANKS with the<br>character's current rank<br>pre-selected<br>specified4ComboBox<br>pre-selects the matching<br>MarineCorp (or "None") | specified3ComboBo<br>x is filled from<br>Marine.RANKS with<br>the character's<br>current rank<br>pre-selected<br>specified4ComboBo<br>x pre-selects the<br>matching<br>MarineCorp (or<br>"None") | P |
|                                                              | 4 | Selecting a Pirate Hunter populates<br>combat style and capture count                       | selectedChar instanceof<br>PirateHunter                                                            | specified1TextField<br>shows combat style<br>specified2TextField<br>shows capture count                                                                                               | specified1TextField<br>shows combat style<br>specified2TextField<br>shows capture<br>count                                                                                                       | P |

|                                             |   |                                                                                            |                                                  |                                                                                                      |                                                                                                      |   |
|---------------------------------------------|---|--------------------------------------------------------------------------------------------|--------------------------------------------------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|---|
|                                             | 5 | Selecting a Civilian populates residence only                                              | selectedChar instanceof Civilian                 | specified1TextField shows residence<br>specified2/3/4 panels remain hidden                           | specified1TextField shows residence<br>specified2/3/4 panels remain hidden                           | P |
|                                             | 6 | Switching from one faction type to another hides the previous type's leftover panels first | Select a Pirate, then select a Civilian          | specified1-4 panels are all set invisible before the Civilian branch re-enables only specified1Panel | specified1-4 panels are all set invisible before the Civilian branch re-enables only specified1Panel | P |
| actionPerformed(ActionEvent) - submitButton | 1 | A field is only committed when its checkbox is checked                                     | nameTextField edited with nameCheckBox unchecked | SetName() is never called while unchecked                                                            | SetName() is never called while unchecked                                                            | P |
|                                             | 2 | A field is only committed when its checkbox is checked                                     | nameTextField edited with nameCheckBox checked   | SetName(nameTextField.getText()) is called with the new value once checked                           | SetName(nameTextField.getText()) is called with the new value once checked                           | P |

|  |   |                                                                                               |                                                  |                                                                                                                                                                   |                                                                                                                                                                   |   |
|--|---|-----------------------------------------------------------------------------------------------|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|  | 3 | Checking the Devil Fruit box with combo box on "None" clears the character's fruit            | dfCheckBox checked, devilfruitComboBox index = 0 | selectedChar.SetDFPower(null) is called                                                                                                                           | selectedChar.SetDFPower(null) is called                                                                                                                           | P |
|  | 4 | Unchecking the Crew box for a Pirate leaves their crew assignment untouched                   | specified4CheckBox unchecked (Pirate)            | SetPirateCrew() is never called even if specified4ComboBox selection differs from the pirate's current crew                                                       | SetPirateCrew() is never called even if specified4ComboBox selection differs from the pirate's current crew                                                       | P |
|  | 5 | Submitting a non-numeric Wallet value with walletCheckBox checked is rejected                 | walletCheckBox checked, walletTextField = "abc"  | NumberFormatException caught dialog reads "Numeric fields (Wallet, Bounty, Captures) must contain valid numbers." no fields are committed after the failure point | NumberFormatException caught dialog reads "Numeric fields (Wallet, Bounty, Captures) must contain valid numbers." no fields are committed after the failure point | P |
|  | 6 | A successful submission shows a confirmation, returns to the menu, and refreshes the dropdown | Valid edits, submitButton clicked                | Success JOptionPane shown panel hidden mainview.showPanel("CHARACTER") called refreshComboBox() is invoked afterward                                              | Success JOptionPane shown panel hidden mainview.showPanel("CHARACTER") called refreshComboBox() is invoked afterward                                              | P |

|                     |   |                                                                                              |                                          |                                                                                                                                 |                                                                                                                                 |   |
|---------------------|---|----------------------------------------------------------------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|---|
|                     | 7 | All four databases are saved regardless of success or failure                                | Submit with a NumberFormatException      | finally block calls writeDatabase() on charDatabase, devilFruitDatabase, pirateCrewDatabase, and marineCorpDatabase.            | finally block calls writeDatabase() on charDatabase, devilFruitDatabase, pirateCrewDatabase, and marineCorpDatabase.            | P |
|                     | 8 | All four databases are saved regardless of success or failure                                | Submit without a NumberFormatException   | finally block calls writeDatabase() on charDatabase, devilFruitDatabase, pirateCrewDatabase, and marineCorpDatabase.            | finally block calls writeDatabase() on charDatabase, devilFruitDatabase, pirateCrewDatabase, and marineCorpDatabase.            | P |
| refreshComboBox()   | 1 | refreshComboBox() repopulates the character list and auto-triggers a load of the first entry | charDatabase contains $\geq 1$ character | charSelectComboBox is cleared and rebuilt as "ID - Name" a simulated selection event populates the form for the first character | charSelectComboBox is cleared and rebuilt as "ID - Name" a simulated selection event populates the form for the first character | P |
| refreshDFComboBox() | 1 | refreshDFComboBox() repopulates the Devil Fruit list with "None" first                       | devilFruitDatabase contains N fruits     | devilfruitComboBox is cleared item 0 is "None" followed by one entry per fruit as "ID - FruitName"                              | devilfruitComboBox is cleared item 0 is "None" followed by one entry per fruit as "ID - FruitName"                              | P |

### Class: CharacterDeleteController

| Method                                                        | # | Test Description                                                         | Sample Input Data                      | Expected Output                                                                                                         | Actual Output                                                                                                           | P/F |
|---------------------------------------------------------------|---|--------------------------------------------------------------------------|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-----|
| actionPerformed(ActionEvent) - backButton                     | 1 | Clicking Back returns to the character menu without deleting anything    | characterDeleteView.backButton clicked | characterDeleteView.setFrame().setVisible(false)<br>mainview.showPanel("CHARACTER") called<br>charDatabase is unchanged | characterDeleteView.setFrame().setVisible(false)<br>mainview.showPanel("CHARACTER") called<br>charDatabase is unchanged | P   |
| actionPerformed(ActionEvent) - deleteButton (selection guard) | 1 | Clicking Delete with nothing selected shows an error and deletes nothing | charSelectComboBox selection = -1      | Error JOptionPane reads "No character selected!"<br>charDatabase size is unchanged                                      | Error JOptionPane reads "No character selected!"<br>charDatabase size is unchanged                                      | P   |

|                                                                               |   |                                                                                             |                                                                      |                                                                             |                                                                             |   |
|-------------------------------------------------------------------------------|---|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------|---|
|                                                                               | 2 | Clicking Delete with a selection index at or beyond the database size is treated as invalid | <code>selectedIndex == charDatabase.getAllCharacters().size()</code> | Treated as out of bounds error dialog shown no deletion occurs              | Treated as out of bounds error dialog shown no deletion occurs              | P |
| <code>actionPerformed(ActionEvent) - deleteButton</code> (confirmation chain) | 1 | Cancelling the first confirmation dialog aborts the deletion                                | Valid selection 1st <code>JOptionPane</code> confirm = NO            | Character remains in <code>charDatabase</code> no further dialogs are shown | Character remains in <code>charDatabase</code> no further dialogs are shown | P |
|                                                                               | 2 | Confirming the first dialog but cancelling the second aborts the deletion                   | 1st confirm = YES, 2nd confirm = NO                                  | Character remains in <code>charDatabase</code>                              | Character remains in <code>charDatabase</code>                              | P |
|                                                                               | 3 | Confirming the first two dialogs but cancelling the third aborts the deletion               | 1st and 2nd confirm = YES, 3rd confirm = NO                          | Character remains in <code>charDatabase</code>                              | Character remains in <code>charDatabase</code>                              | P |

|                                                                     |   |                                                                         |                                      |                                                                                                        |                                                                                                        |   |
|---------------------------------------------------------------------|---|-------------------------------------------------------------------------|--------------------------------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|---|
|                                                                     | 4 | Confirming all three dialogs deletes the selected character             | All three confirms = YES             | charDatabase.getAllCharacters().remove(selectedIndex) is called the character is no longer in the list | charDatabase.getAllCharacters().remove(selectedIndex) is called the character is no longer in the list | P |
| actionPerformed(ActionEvent) - deleteButton (post-deletion effects) | 1 | A successful deletion shows a confirmation naming the removed character | All three confirms = YES for "Luffy" | JOptionPane message reads "Luffy has been successfully unalived."                                      | JOptionPane message reads "Luffy has been successfully unalived."                                      | P |
|                                                                     | 2 | A successful deletion refreshes the character dropdown                  | All three confirms = YES             | refreshComboBox() is invoked, removing the deleted character from charSelectComboBox                   | refreshComboBox() is invoked, removing the deleted character from charSelectComboBox                   | P |
|                                                                     | 3 | A successful deletion returns to the character menu                     | All three confirms = YES             | characterDeleteView.setFrame().setVisible(false) mainview.showPanel("CHARACTER") called                | characterDeleteView.setFrame().setVisible(false) mainview.showPanel("CHARACTER") called                | P |

|                   |   |                                                                                              |                                                                                          |                                                                                                                                                                            |                                                                                                                                                                            |   |
|-------------------|---|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|                   | 4 | The database is saved on every deleteButton click, success or not                            | deleteButton clicked with no selection, or with a cancelled/completed confirmation chain | charDatabase.writeDatabase() is called at the end of the branch in every case                                                                                              | charDatabase.writeDatabase() is called at the end of the branch in every case                                                                                              | P |
|                   | 5 | Cancelling anywhere in the confirmation chain does not navigate away or refresh the dropdown | Any of the three confirm dialogs answered NO                                             | characterDeleteView frame stays visible, mainview.showPanel() is not called, and refreshComboBox() is not invoked                                                          | characterDeleteView frame stays visible, mainview.showPanel() is not called, and refreshComboBox() is not invoked                                                          | P |
| refreshComboBox() | 1 | Refreshing with characters present repopulates the dropdown and simulates a selection        | charDatabase contains $\geq 1$ character<br>refreshComboBox() called                     | charSelectComboBox is cleared and rebuilt as "ID - Name" listener is detached/reattached around the rebuild a simulated ACTION_PERFORMED event is fired for the first item | charSelectComboBox is cleared and rebuilt as "ID - Name" listener is detached/reattached around the rebuild a simulated ACTION_PERFORMED event is fired for the first item | P |
|                   | 2 | Refreshing after the last character is deleted leaves the dropdown empty                     | charDatabase becomes empty after deletion<br>refreshComboBox() called                    | charSelectComboBox ends up empty no simulated event is fired and no exception is thrown                                                                                    | charSelectComboBox ends up empty no simulated event is fired and no exception is thrown                                                                                    | P |

|  |   |                                                                                       |                                                                                      |                                                                                          |                                                                                          |   |
|--|---|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|---|
|  | 3 | Deleting a character other than the first item still deletes the correct one by index | 3 characters in charDatabase user selects and confirms deletion of the 2nd (index 1) | Only the character originally at index 1 is removed the other two remain in charDatabase | Only the character originally at index 1 is removed the other two remain in charDatabase | P |
|--|---|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|---|

### Class: DevilFruitViewController

| Method                                                       | # | Test Description                                                       | Sample Input Data                                     | Expected Output                                                                                                                | Actual Output                                                                                                                  | P/F |
|--------------------------------------------------------------|---|------------------------------------------------------------------------|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-----|
| <code>actionPerformed(ActionEvent) - backButton</code>       | 1 | Clicking Back hides the view panel and returns to the Devil Fruit menu | <code>devilFruitViewView.backButton</code> clicked    | <code>devilFruitViewView.setFrame().setVisible(false)</code><br><code>mainview.showPanel("DEVILFRUIT")</code> called           | <code>devilFruitViewView.setFrame().setVisible(false)</code><br><code>mainview.showPanel("DEVILFRUIT")</code> called           | P   |
| <code>actionPerformed(ActionEvent) - dfSelectComboBox</code> | 1 | Selecting a valid fruit updates the name, category, and ability labels | <code>dfSelectComboBox</code> selection = valid index | <code>nameLabel</code> , <code>categoryLabel</code> , and <code>abilityLabel</code> are updated with the selected fruit's data | <code>nameLabel</code> , <code>categoryLabel</code> , and <code>abilityLabel</code> are updated with the selected fruit's data | P   |

|  |   |                                                                       |                                                                                |                                                                                                         |                                                                                                         |   |
|--|---|-----------------------------------------------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|---|
|  | 2 | Selecting a fruit with no current owner shows "None"                  | <code>selectedFruit.HasCurrentOwner() == false</code>                          | currentOwnerLabel text reads "Current Owner: None"                                                      | currentOwnerLabel text reads "Current Owner: None"                                                      | P |
|  | 3 | Selecting a fruit with a current owner shows the owner's name         | <code>selectedFruit.HasCurrentOwner() == true</code>                           | currentOwnerLabel text reads "Current Owner: " + <code>selectedFruit.GetCurrentOwner().GetName()</code> | currentOwnerLabel text reads "Current Owner: " + <code>selectedFruit.GetCurrentOwner().GetName()</code> | P |
|  | 4 | Selecting a fruit passes its historical owners list to the view       | <code>selectedFruit.GetHistoricalOwners()</code> returns a list of past owners | <code>devilFruitViewView.setHistoricalOwners()</code> is called with exactly that list                  | <code>devilFruitViewView.setHistoricalOwners()</code> is called with exactly that list                  | P |
|  | 5 | Selecting a fruit with an empty historical owners list does not error | <code>selectedFruit.GetHistoricalOwners()</code> returns an empty list         | <code>setHistoricalOwners()</code> is called with an empty list no exception is thrown                  | <code>setHistoricalOwners()</code> is called with an empty list no exception is thrown                  | P |

|                        |   |                                                                                          |                                                                                        |                                                                                                                                                  |                                                                                                                                                  |   |
|------------------------|---|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|---|
|                        | 6 | Selecting a combo box index outside the database bounds is ignored                       | dfSelectComboBox selection = -1 (or $\geq$ devilFruitDatabase size)                    | No labels are updated and no exception is thrown                                                                                                 | No labels are updated and no exception is thrown                                                                                                 | P |
|                        | 7 | An ActionEvent from an unrelated source is ignored                                       | actionPerformed() called with a source that is neither backButton nor dfSelectComboBox | No labels change and no exception is thrown                                                                                                      | No labels change and no exception is thrown                                                                                                      | P |
| refreshComboBox()<br>( | 1 | Refreshing with fruits present repopulates the dropdown and auto-selects the first entry | devilFruitDatabase contains $\geq 1$ fruit<br>refreshComboBox() called                 | dfSelectComboBox is cleared and repopulated as "ID - FruitName" a simulated ACTION_PERFORMED event fires, updating the labels for the first item | dfSelectComboBox is cleared and repopulated as "ID - FruitName" a simulated ACTION_PERFORMED event fires, updating the labels for the first item | P |
|                        | 2 | Refreshing an empty database clears the dropdown without simulating a selection          | devilFruitDatabase is empty<br>refreshComboBox() called                                | dfSelectComboBox ends up empty no simulated ActionEvent is fired and no exception is thrown                                                      | dfSelectComboBox ends up empty no simulated ActionEvent is fired and no exception is thrown                                                      | P |

|                                                  |   |                                                                                                                                  |                                                                   |                                                                                                                                                               |                                                                                                                                                               |   |
|--------------------------------------------------|---|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|                                                  | 3 | The action listener is detached before the rebuild and reattached after                                                          | refreshComboBox() called while dfSelectComboBox already has items | removeActionListener() is called before clearing the box, and addActionListener() is called again after repopulating, preventing a listener storm mid-rebuild | removeActionListener() is called before clearing the box, and addActionListener() is called again after repopulating, preventing a listener storm mid-rebuild | P |
| actionPerformed(ActionEvent) - selection changes | 1 | Selecting a different fruit after one is already displayed replaces all label text with no stale data left over                  | Select Fruit A, then select Fruit B                               | All five labels (name, category, ability, current owner, historical owners) reflect Fruit B only none retain Fruit A's values                                 | All five labels (name, category, ability, current owner, historical owners) reflect Fruit B only none retain Fruit A's values                                 | P |
|                                                  | 2 | refreshComboBox() called while a selection is active resets display to the first item rather than preserving the prior selection | dfSelectComboBox already showing Fruit B refreshComboBox() called | The combo box and all labels reset to reflect the first fruit in the rebuilt list, not the previously selected Fruit B                                        | The combo box and all labels reset to reflect the first fruit in the rebuilt list, not the previously selected Fruit B                                        | P |

### Class: DevilFruitAssignController

| Method                                                   | # | Test Description                                                          | Sample Input Data                                                                              | Expected Output                                                                                                                       | Actual Output                                                                                                                         | P/F |
|----------------------------------------------------------|---|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-----|
| <code>actionPerformed(ActionEvent) - exitButton</code>   | 1 | Clicking Exit discards the in-progress assignment and returns to the menu | <code>devilFruitAssignView.exitButton</code> clicked                                           | <code>devilFruitAssignView.setFrame().setVisible(false)</code><br><code>mainview.showPanel("DEVILFRUIT")</code> called                | <code>devilFruitAssignView.setFrame().setVisible(false)</code><br><code>mainview.showPanel("DEVILFRUIT")</code> called                | P   |
| <code>actionPerformed(ActionEvent) - submitButton</code> | 1 | Submitting with both a fruit and a character selected assigns ownership   | <code>dfComboBox</code> and <code>charComboBox</code> both have a valid ( $\geq 0$ ) selection | <code>selectedDF.SetCurrentOwner(selectedChar)</code> is called with the objects fetched from their databases at the selected indices | <code>selectedDF.SetCurrentOwner(selectedChar)</code> is called with the objects fetched from their databases at the selected indices | P   |

|  |   |                                                                                      |                                                           |                                                                                                                         |                                                                                                                         |   |
|--|---|--------------------------------------------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|---|
|  | 2 | A successful assignment shows a confirmation naming both the fruit and the character | Valid selections, ex: "Gomu Gomu no Mi" and "Luffy"       | JOptionPane message reads "Gomu Gomu no Mi has been assigned to Luffy!"                                                 | JOptionPane message reads "Gomu Gomu no Mi has been assigned to Luffy!"                                                 | P |
|  | 3 | A successful assignment closes the panel and returns to the menu                     | Valid selections                                          | devilFruitAssignView setFrame().setVisible(false) mainview.showPanel("DEVILFRUIT") called                               | devilFruitAssignView setFrame().setVisible(false) mainview.showPanel("DEVILFRUIT") called                               | P |
|  | 4 | Submitting with no Devil Fruit selected is rejected                                  | dfComboBox selection = -1, charComboBox selection = valid | Warning dialog reads "Please ensure both a Devil Fruit and a Character are selected." SetCurrentOwner() is never called | Warning dialog reads "Please ensure both a Devil Fruit and a Character are selected." SetCurrentOwner() is never called | P |
|  | 5 | Submitting with no Character selected is rejected                                    | charComboBox selection = -1, dfComboBox selection = valid | Warning dialog reads "Please ensure both a Devil Fruit and a Character are selected." SetCurrentOwner() is never called | Warning dialog reads "Please ensure both a Devil Fruit and a Character are selected." SetCurrentOwner() is never called | P |

|                     |   |                                                                                                            |                                                                                                     |                                                                                                                       |                                                                                                                       |   |
|---------------------|---|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---|
|                     | 6 | An exception thrown during assignment is caught and reported                                               | selectedDF.SetCurrentOwner() throws an exception (ex: fruit already owned)                          | Error dialog reads "Error assigning Devil Fruit: " + the exception's message the panel does not close                 | Error dialog reads "Error assigning Devil Fruit: " + the exception's message the panel does not close                 | P |
|                     | 7 | Both databases are saved after every submit click, regardless of outcome                                   | Submit with a valid assignment, a missing selection, and a thrown exception (three separate clicks) | charDatabase.writeDatabase() and devilFruitDatabase.writeDatabase() are each called once per click in all three cases | charDatabase.writeDatabase() and devilFruitDatabase.writeDatabase() are each called once per click in all three cases | P |
|                     | 8 | Re-submitting a new fruit/character pair after a prior successful assignment invokes SetCurrentOwner again | Submit and confirm Fruit A → Character A, then reopen and submit Fruit B → Character B              | SetCurrentOwner() is called once per submit click, correctly reflecting each new pairing independently                | SetCurrentOwner() is called once per submit click, correctly reflecting each new pairing independently                | P |
| refreshComboBoxes() | 1 | Refreshing repopulates the Devil Fruit dropdown from the database                                          | devilFruitDatabase contains N fruits<br>refreshComboBoxes() called                                  | dfComboBox is cleared and repopulated with one entry per fruit as "ID - FruitName"                                    | dfComboBox is cleared and repopulated with one entry per fruit as "ID - FruitName"                                    | P |

|  |   |                                                                          |                                                                  |                                                                                     |                                                                                     |   |
|--|---|--------------------------------------------------------------------------|------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---|
|  | 2 | Refreshing repopulates the Character dropdown from the database          | charDatabase contains N characters<br>refreshComboBoxes() called | charComboBox is cleared and repopulated with one entry per character as "ID - Name" | charComboBox is cleared and repopulated with one entry per character as "ID - Name" | P |
|  | 3 | Refreshing with an empty Devil Fruit database leaves that dropdown empty | devilFruitDatabase is empty<br>refreshComboBoxes() called        | dfComboBox ends up empty no exception is thrown                                     | dfComboBox ends up empty no exception is thrown                                     | P |
|  | 4 | Refreshing with an empty Character database leaves that dropdown empty   | charDatabase is empty<br>refreshComboBoxes() called              | charComboBox ends up empty no exception is thrown                                   | charComboBox ends up empty no exception is thrown                                   | P |

### Class: DevilFruitController

| Method                                                   | # | Test Description                                                        | Sample Input Data                                | Expected Output                                                                                                                  | Actual Output                                                                                                                    | P/F |
|----------------------------------------------------------|---|-------------------------------------------------------------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-----|
| <code>actionPerformed(ActionEvent) - createButton</code> | 1 | Clicking Create sets info text and switches to the create panel         | <code>devilFruitView.createButton</code> clicked | <code>mainview.setInfoText("Creating a Devil Fruit")</code> called<br><code>mainview.showPanel("DEVILFRUITCREATE")</code> called | <code>mainview.setInfoText("Creating a Devil Fruit")</code> called<br><code>mainview.showPanel("DEVILFRUITCREATE")</code> called | P   |
|                                                          | 2 | Clicking Create does not refresh any combo box before showing the panel | <code>devilFruitView.createButton</code> clicked | Unlike the view/assign branches, no refresh method on <code>devilFruitCreateController</code> is invoked (none exists to call)   | Unlike the view/assign branches, no refresh method on <code>devilFruitCreateController</code> is invoked (none exists to call)   | P   |

|                                             |   |                                                                   |                                     |                                                                                           |                                                                                           |   |
|---------------------------------------------|---|-------------------------------------------------------------------|-------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|---|
| actionPerformed(ActionEvent) - viewButton   | 1 | Clicking View sets info text and switches to the view panel       | devilFruitView.viewButton clicked   | setInfoText("Viewing a Devil Fruit") called<br>showPanel("DEVILFRUITVIEW") called         | setInfoText("Viewing a Devil Fruit") called<br>showPanel("DEVILFRUITVIEW") called         | P |
|                                             | 2 | Clicking View refreshes the Devil Fruit selection combo box first | devilFruitView.viewButton clicked   | devilFruitViewController.refreshComboBox() is invoked before<br>showPanel() is called     | devilFruitViewController.refreshComboBox() is invoked before<br>showPanel() is called     | P |
| actionPerformed(ActionEvent) - assignButton | 1 | Clicking Assign sets info text and switches to the assign panel   | devilFruitView.assignButton clicked | setInfoText("Assigning a Devil Fruit") called<br>showPanel("DEVILFRUITASSIGN") called     | setInfoText("Assigning a Devil Fruit") called<br>showPanel("DEVILFRUITASSIGN") called     | P |
|                                             | 2 | Clicking Assign refreshes both combo boxes first                  | devilFruitView.assignButton clicked | devilFruitAssignController.refreshComboBoxes() is invoked before<br>showPanel() is called | devilFruitAssignController.refreshComboBoxes() is invoked before<br>showPanel() is called | P |

|                                                 |   |                                                                                   |                                                               |                                                                                                                                           |                                                                                                                                           |   |
|-------------------------------------------------|---|-----------------------------------------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|---|
| actionPerformed(ActionEvent) - backButton       | 1 | Clicking Back hides the Devil Fruit panel and resets info text without navigating | devilFruitView.backButton clicked                             | setInfoText("Main Menu") called<br>devilFruitView.panel.setVisible(false) no<br>mainview.showPanel() call is made                         | setInfoText("Main Menu") called<br>devilFruitView.panel.setVisible(false) no<br>mainview.showPanel() call is made                         | P |
| actionPerformed(ActionEvent) - unmatched source | 2 | An ActionEvent from a source not registered to this controller is ignored         | actionPerformed() called with an unrelated ActionEvent source | No info text change, no panel switch, and no exception is thrown                                                                          | No info text change, no panel switch, and no exception is thrown                                                                          | P |
|                                                 | 3 | Clicking Create then View in sequence updates info text correctly each time       | createButton clicked, then viewButton clicked                 | Info text reads "Creating a Devil Fruit" after the first click and "Viewing a Devil Fruit" after the second, with no stale text left over | Info text reads "Creating a Devil Fruit" after the first click and "Viewing a Devil Fruit" after the second, with no stale text left over | P |

### Class: DevilFruitCreateController

| Method                                                   | # | Test Description                                                             | Sample Input Data                                               | Expected Output                                                                                                   | Actual Output                                                                                                     | P/F |
|----------------------------------------------------------|---|------------------------------------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-----|
| <code>actionPerformed(ActionEvent) - exitButton</code>   | 1 | Clicking Exit discards in-progress input and returns to the Devil Fruit menu | <code>devilFruitCreateView.exitButton</code> clicked            | <code>devilFruitCreateView.panel.setVisible(false)</code><br><code>mainview.showPanel("DEVILFRUIT")</code> called | <code>devilFruitCreateView.panel.setVisible(false)</code><br><code>mainview.showPanel("DEVILFRUIT")</code> called | P   |
| <code>actionPerformed(ActionEvent) - submitButton</code> | 1 | Submitting with Paramecia selected creates a fruit of that category          | <code>parameciaRadioButton</code> selected, Name/Ability filled | New DevilFruit is constructed with category = "Paramecia"                                                         | New DevilFruit is constructed with category = "Paramecia"                                                         | P   |

|  |   |                                                                                |                                                |                                                                                                                     |                                                                                                                     |   |
|--|---|--------------------------------------------------------------------------------|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|---|
|  | 2 | Submitting with Zoan selected creates a fruit of that category                 | zoanRadioButton selected, Name/Ability filled  | New DevilFruit is constructed with category = "Zoan"                                                                | New DevilFruit is constructed with category = "Zoan"                                                                | P |
|  | 3 | Submitting with Logia selected creates a fruit of that category                | logiaRadioButton selected, Name/Ability filled | New DevilFruit is constructed with category = "Logia"                                                               | New DevilFruit is constructed with category = "Logia"                                                               | P |
|  | 4 | Submitting with no category radio button selected is not blocked by validation | No radio button selected, Name/Ability filled  | category remains "" the fruit is still constructed and added with an empty category string (no exception is thrown) | category remains "" the fruit is still constructed and added with an empty category string (no exception is thrown) | P |
|  | 5 | Submitting with a blank Name field is rejected                                 | nameTextField = "", abilityTextField filled    | EmptyInputException caught error dialog reads "Error: Name cannot be empty" no fruit is added                       | EmptyInputException caught error dialog reads "Error: Name cannot be empty" no fruit is added                       | P |

|  |   |                                                                              |                                                      |                                                                                                               |                                                                                                               |   |
|--|---|------------------------------------------------------------------------------|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|---|
|  | 6 | Submitting with a blank Ability field is rejected                            | abilityTextField = "",<br>nameTextField filled       | EmptyInputException caught error dialog reads "Error: Ability cannot be empty" no fruit is added              | EmptyInputException caught error dialog reads "Error: Ability cannot be empty" no fruit is added              | P |
|  | 7 | A successful submission constructs the fruit with no owner and empty history | Valid Name/Ability/Category                          | new DevilFruit(name, category, ability, null, new ArrayList<>()) is added to devilFruitDatabase.getAlIDF()    | new DevilFruit(name, category, ability, null, new ArrayList<>()) is added to devilFruitDatabase.getAlIDF()    | P |
|  | 8 | A successful submission logs a confirmation to the console                   | Valid submission for a fruit named "Mera Mera no Mi" | Console prints "Successfully added Mera Mera no Mi to the Module 3 Database."                                 | Console prints "Successfully added Mera Mera no Mi to the Module 3 Database."                                 | P |
|  | 9 | A successful submission shows a success dialog and returns to the menu       | Valid submission                                     | JOptionPane reads "Devil Fruit Created Successfully!" panel hidden<br>mainview.showPanel("DEVILFRUIT") called | JOptionPane reads "Devil Fruit Created Successfully!" panel hidden<br>mainview.showPanel("DEVILFRUIT") called | P |

|                                                       |    |                                                                              |                                                                                   |                                                                                                                                                         |                                                                                                                                                         |   |
|-------------------------------------------------------|----|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|                                                       | 10 | The database is saved on submit whether it succeeds or throws                | Submit once with valid data, once with a blank Name                               | finally block calls devilFruitDatabase.writeDatabase() in both cases                                                                                    | finally block calls devilFruitDatabase.writeDatabase() in both cases                                                                                    | P |
| actionPerformed(ActionEvent) - category radio buttons | 1  | Clicking a category radio button does not itself throw or change any label   | parameciaRadioButton, zoanRadioButton, or logiaRadioButton clicked directly       | actionPerformed() runs but matches no explicit if/else branch for these sources no exception occurs and no output changes until submitButton is clicked | actionPerformed() runs but matches no explicit if/else branch for these sources no exception occurs and no output changes until submitButton is clicked | P |
| actionPerformed(ActionEvent) - repeated submissions   | 1  | Two consecutive successful submissions create two distinct database entries  | Submit Fruit A, then submit Fruit B with different Name/Ability values            | devilFruitDatabase.getAllIDF() contains two separate DevilFruit objects, one for each submission                                                        | devilFruitDatabase.getAllIDF() contains two separate DevilFruit objects, one for each submission                                                        | P |
|                                                       | 2  | Only the field values present at the moment submitButton is clicked are used | Change nameTextField's text after selecting a category but before clicking submit | The created DevilFruit uses the final text in nameTextField/abilityTextField at click time, not any earlier value                                       | The created DevilFruit uses the final text in nameTextField/abilityTextField at click time, not any earlier value                                       | P |

## Class: MarineCorpController

| Method                                    | # | Test Description                  | Sample Input Data                             | Expected Output                                          | Actual Output                                            | P/F |
|-------------------------------------------|---|-----------------------------------|-----------------------------------------------|----------------------------------------------------------|----------------------------------------------------------|-----|
| <code>addActionListener()</code>          | 1 | All buttons respond to clicks     | 1. Click each of the four buttons one by one. | Each button click leads to its correct screen or action. | Each button click leads to its correct screen or action. | P   |
| <code>actionPerformed(ActionEvent)</code> | 1 | Create button opens Create screen | 1. Click the Create button.                   | Marine Corp Create screen is shown.                      | Marine Corp Create screen is shown.                      | P   |
|                                           | 2 | Create button updates info text   | 1. Click the Create button.                   | Info text changes to 'Marine Corp Create'.               | Info text changes to 'Marine Corp Create'.               | P   |

|  |   |                                     |                                                        |                                                         |                                                         |   |
|--|---|-------------------------------------|--------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------|---|
|  | 3 | Create button refreshes marine list | 1. Add a new Marine.<br>2. Click the Create button.    | Create screen's commander list includes the new Marine. | Create screen's commander list includes the new Marine. | P |
|  | 4 | View button opens View screen       | 1. Click the View button.                              | Marine Corp View screen is shown.                       | Marine Corp View screen is shown.                       | P |
|  | 5 | View button refreshes corp list     | 1. Add a new Marine Corp.<br>2. Click the View button. | View screen's combo box includes the new corp.          | View screen's combo box includes the new corp.          | P |
|  | 6 | Edit button opens Modify screen     | 1. Click the Edit button.                              | Marine Corp Modify screen is shown.                     | Marine Corp Modify screen is shown.                     | P |

|  |    |                                      |                                                        |                                                         |                                                         |   |
|--|----|--------------------------------------|--------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------|---|
|  | 7  | Edit button refreshes corp list      | 1. Add a new Marine Corp.<br>2. Click the Edit button. | Modify screen's combo box includes the new corp.        | Modify screen's combo box includes the new corp.        | P |
|  | 8  | Back button hides the menu           | 1. Click the Back button.                              | Marine Corp panel becomes hidden.                       | Marine Corp panel becomes hidden.                       | P |
|  | 9  | Back button updates info text        | 1. Click the Back button.                              | Info text changes to 'Main Menu'.                       | Info text changes to 'Main Menu'.                       | P |
|  | 10 | Button sequence works with no errors | 1. Click Create.<br>2. Click Back.<br>3. Click View.   | Screens change correctly at each step, with no crashes. | Screens change correctly at each step, with no crashes. | P |

|  |    |                              |                                                      |                           |                           |   |
|--|----|------------------------------|------------------------------------------------------|---------------------------|---------------------------|---|
|  | 11 | Unrelated click does nothing | 1. Trigger an action event from an unrelated source. | No screen change happens. | No screen change happens. | P |
|--|----|------------------------------|------------------------------------------------------|---------------------------|---------------------------|---|

### Class: MarineCorpCreateController

| Method        | # | Test Description                        | Sample Input Data                     | Expected Output                                      | Actual Output                                        | P/F |
|---------------|---|-----------------------------------------|---------------------------------------|------------------------------------------------------|------------------------------------------------------|-----|
| refreshData() | 1 | Commander list is cleared before reload | 1. Call refreshData() twice in a row. | Commander combo box does not show duplicate entries. | Commander combo box does not show duplicate entries. | P   |
|               | 2 | 'None' is the first commander choice    | 1. Call refreshData().                | First item in commander combo box is 'None'.         | First item in commander combo box is 'None'.         | P   |

|                              |   |                                      |                                                                          |                                                                    |                                                                    |   |
|------------------------------|---|--------------------------------------|--------------------------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------|---|
|                              | 3 | Only Marines are listed              | 1. Add both Marine and non-Marine characters.<br>2. Call refreshData().  | Only Marine characters appear in the commander list and checklist. | Only Marine characters appear in the commander list and checklist. | P |
|                              | 4 | Checklist shows one row per Marine   | 1. Add three Marines.<br>2. Call refreshData().                          | Checklist panel shows three selectable rows.                       | Checklist panel shows three selectable rows.                       | P |
| actionPerformed(ActionEvent) | 1 | Create with valid location and funds | 1. Enter a location.<br>2. Enter a number for funds.<br>3. Click Submit. | New Marine Corp is saved to the database.                          | New Marine Corp is saved to the database.                          | P |
|                              | 2 | Create with a chosen commander       | 1. Pick a Marine as commander.<br>2. Click Submit.                       | New corp's commander is set to that Marine.                        | New corp's commander is set to that Marine.                        | P |

|  |   |                                 |                                                            |                                                      |                                                      |   |
|--|---|---------------------------------|------------------------------------------------------------|------------------------------------------------------|------------------------------------------------------|---|
|  | 3 | Chosen commander flag updates   | 1. Pick a Marine as commander.<br>2. Click Submit.         | That Marine's 'is corp commander' flag becomes true. | That Marine's 'is corp commander' flag becomes true. | P |
|  | 4 | Create with 'None' as commander | 1. Leave commander as 'None'.<br>2. Click Submit.          | New corp has no commander set.                       | New corp has no commander set.                       | P |
|  | 5 | Checked members are added       | 1. Check two Marines in the checklist.<br>2. Click Submit. | Both Marines are added as corp members.              | Both Marines are added as corp members.              | P |
|  | 6 | Funds field with letters        | 1. Type letters into the funds field.<br>2. Click Submit.  | Error message says funds must be a valid number.     | Error message says funds must be a valid number.     | P |

|  |    |                                       |                                                      |                                                       |                                                       |   |
|--|----|---------------------------------------|------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------|---|
|  | 7  | Successful create message             | 1. Fill the form correctly.<br>2. Click Submit.      | 'Marine Corp Created Successfully!' message is shown. | 'Marine Corp Created Successfully!' message is shown. | P |
|  | 8  | Screen closes after successful create | 1. Fill the form correctly.<br>2. Click Submit.      | Create screen closes and Marine Corp menu is shown.   | Create screen closes and Marine Corp menu is shown.   | P |
|  | 9  | Exit button cancels the form          | 1. Click the Exit button.                            | Create screen closes with no new corp saved.          | Create screen closes with no new corp saved.          | P |
|  | 10 | Database is saved after submit        | 1. Click Submit, whether it works or shows an error. | Character and Marine Corp databases are both saved.   | Character and Marine Corp databases are both saved.   | P |

|  |    |                            |                                                                                    |                                                    |                                                    |   |
|--|----|----------------------------|------------------------------------------------------------------------------------|----------------------------------------------------|----------------------------------------------------|---|
|  | 11 | Create with blank location | 1. Leave location field blank.<br>2. Fill funds with a number.<br>3. Click Submit. | New corp is still created, with an empty location. | New corp is still created, with an empty location. | P |
|--|----|----------------------------|------------------------------------------------------------------------------------|----------------------------------------------------|----------------------------------------------------|---|

### Class: MarineCorpModifyController

| Method                       | # | Test Description                  | Sample Input Data                    | Expected Output                                          | Actual Output                                            | P/F |
|------------------------------|---|-----------------------------------|--------------------------------------|----------------------------------------------------------|----------------------------------------------------------|-----|
| actionPerformed(ActionEvent) | 1 | Selecting a corp fills the form   | 1. Pick a corp from the combo box.   | Location and funds fields fill in with that corp's data. | Location and funds fields fill in with that corp's data. | P   |
|                              | 2 | Selecting a corp with a commander | 1. Pick a corp that has a commander. | Commander combo box shows the right commander selected.  | Commander combo box shows the right commander selected.  | P   |

|  |   |                                    |                                                                                     |                                                             |                                                             |   |
|--|---|------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------|---|
|  | 3 | Selecting a corp with no commander | 1. Pick a corp with no commander.                                                   | Commander combo box shows 'None' selected.                  | Commander combo box shows 'None' selected.                  | P |
|  | 4 | Member checkboxes match the corp   | 1. Pick a corp with known members.                                                  | Checklist shows those members checked and others unchecked. | Checklist shows those members checked and others unchecked. | P |
|  | 5 | Update location                    | 1. Check the location checkbox.<br>2. Change the location text.<br>3. Click Submit. | Corp's location is updated to the new text.                 | Corp's location is updated to the new text.                 | P |
|  | 6 | Location box left unchecked        | 1. Change the location text but leave the checkbox unchecked.<br>2. Click Submit.   | Corp's location does not change.                            | Corp's location does not change.                            | P |

|  |    |                          |                                                                                                                                                              |                                                  |                                                  |   |
|--|----|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|--------------------------------------------------|---|
|  | 7  | Update funds             | <ol style="list-style-type: none"> <li>1. Check the funds checkbox.</li> <li>2. Enter a new number.</li> <li>3. Click Submit.</li> </ol>                     | Corp's funds are updated to the new number.      | Corp's funds are updated to the new number.      | P |
|  | 8  | Funds field with letters | <ol style="list-style-type: none"> <li>1. Check the funds checkbox.</li> <li>2. Type letters instead of numbers.</li> <li>3. Click Submit.</li> </ol>        | Error message says funds must be a valid number. | Error message says funds must be a valid number. | P |
|  | 9  | Update commander         | <ol style="list-style-type: none"> <li>1. Check the commander checkbox.</li> <li>2. Pick a new commander.</li> <li>3. Click Submit.</li> </ol>               | Corp's commander changes to the new pick.        | Corp's commander changes to the new pick.        | P |
|  | 10 | Add a member             | <ol style="list-style-type: none"> <li>1. Check the members checkbox.</li> <li>2. Check a Marine that was not a member.</li> <li>3. Click Submit.</li> </ol> | That Marine is added to the corp.                | That Marine is added to the corp.                | P |

|                         |    |                                   |                                                                                                                                                            |                                                        |                                                        |   |
|-------------------------|----|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------|---|
|                         | 11 | Remove a member                   | <ol style="list-style-type: none"> <li>1. Check the members checkbox.</li> <li>2. Uncheck a Marine that was a member.</li> <li>3. Click Submit.</li> </ol> | That Marine is removed from the corp.                  | That Marine is removed from the corp.                  | P |
|                         | 12 | Successful modify message         | <ol style="list-style-type: none"> <li>1. Make a change.</li> <li>2. Click Submit.</li> </ol>                                                              | 'Marine Corp Modified Successfully!' message is shown. | 'Marine Corp Modified Successfully!' message is shown. | P |
|                         | 13 | Exit button cancels changes       | <ol style="list-style-type: none"> <li>1. Click the Exit button.</li> </ol>                                                                                | Modify screen closes and Marine Corp menu is shown.    | Modify screen closes and Marine Corp menu is shown.    | P |
| refreshComboBox()<br>() | 1  | Refresh with no corps in database | <ol style="list-style-type: none"> <li>1. Remove all corps.</li> <li>2. Call refreshComboBox().</li> </ol>                                                 | All fields and the checklist are cleared.              | All fields and the checklist are cleared.              | P |

|  |   |                                 |                                                    |                                                      |                                                      |   |
|--|---|---------------------------------|----------------------------------------------------|------------------------------------------------------|------------------------------------------------------|---|
|  | 2 | Refresh reloads the Marine list | 1. Add a new Marine.<br>2. Call refreshComboBox(). | Commander list and checklist include the new Marine. | Commander list and checklist include the new Marine. | P |
|--|---|---------------------------------|----------------------------------------------------|------------------------------------------------------|------------------------------------------------------|---|

### Class: MarineCorpViewController

| Method                       | # | Test Description                          | Sample Input Data                  | Expected Output                                          | Actual Output                                            | P/F |
|------------------------------|---|-------------------------------------------|------------------------------------|----------------------------------------------------------|----------------------------------------------------------|-----|
| actionPerformed(ActionEvent) | 1 | Back button goes back to Marine Corp menu | 1. Click the Back button.          | Screen changes to the Marine Corp menu.                  | Screen changes to the Marine Corp menu.                  | P   |
|                              | 2 | Selecting a corp shows its location       | 1. Pick a corp from the combo box. | Location label shows 'Marine Corp Location: <location>'. | Location label shows 'Marine Corp Location: <location>'. | P   |

|  |   |                                    |                                      |                                                   |                                                   |   |
|--|---|------------------------------------|--------------------------------------|---------------------------------------------------|---------------------------------------------------|---|
|  | 3 | Selecting a corp shows its funds   | 1. Pick a corp from the combo box.   | Funds label shows 'Operational Funds: <amount>'.  | Funds label shows 'Operational Funds: <amount>'.  | P |
|  | 4 | Selecting a corp with a commander  | 1. Pick a corp that has a commander. | Commander label shows 'Commander's Name: <name>'. | Commander label shows 'Commander's Name: <name>'. | P |
|  | 5 | Selecting a corp with no commander | 1. Pick a corp with no commander.    | Commander label shows 'Commander's Name: None'.   | Commander label shows 'Commander's Name: None'.   | P |
|  | 6 | Member panels show up correctly    | 1. Pick a corp that has members.     | Inner panel shows one panel for each member.      | Inner panel shows one panel for each member.      | P |

|                         |   |                                       |                                                     |                                                      |                                                      |   |
|-------------------------|---|---------------------------------------|-----------------------------------------------------|------------------------------------------------------|------------------------------------------------------|---|
|                         | 7 | Switching between two corps           | 1. Select Corp A.<br>2. Select Corp B.              | Labels update to show Corp B's details.              | Labels update to show Corp B's details.              | P |
|                         | 8 | Selecting the same corp again         | 1. Select a corp.<br>2. Select the same corp again. | Labels stay correct with no error.                   | Labels stay correct with no error.                   | P |
| refreshComboBox()<br>() | 1 | Refresh combo box picks first corp    | 1. Call refreshComboBox()<br>while corps exist.     | First corp is auto-selected and its details show up. | First corp is auto-selected and its details show up. | P |
|                         | 2 | Refresh combo box with empty database | 1. Remove all corps.<br>2. Call refreshComboBox().  | All labels are cleared.                              | All labels are cleared.                              | P |

|  |   |                                                      |                                                        |                                                      |                                                      |   |
|--|---|------------------------------------------------------|--------------------------------------------------------|------------------------------------------------------|------------------------------------------------------|---|
|  | 3 | Refresh combo box with empty database clears members | 1. Remove all corps.<br>2. Call refreshComboBox().     | Inner panel has no member panels left.               | Inner panel has no member panels left.               | P |
|  | 4 | New corp shows up after refresh                      | 1. Add a new corp.<br>2. Call refreshComboBox().       | New corp appears in the combo box list.              | New corp appears in the combo box list.              | P |
|  | 5 | Combo box text format is correct                     | 1. Open the View screen.                               | Each combo box entry shows '<corp ID> - <location>'. | Each combo box entry shows '<corp ID> - <location>'. | P |
|  | 6 | View shows the latest changes                        | 1. Modify a corp's data.<br>2. Reopen the View screen. | View screen shows the updated data.                  | View screen shows the updated data.                  | P |

## Class: PirateCrewController

| Method                                    | # | Test Description                  | Sample Input Data                             | Expected Output                                          | Actual Output | P/F |
|-------------------------------------------|---|-----------------------------------|-----------------------------------------------|----------------------------------------------------------|---------------|-----|
| <code>addActionListener()</code>          | 1 | All buttons respond to clicks     | 1. Click each of the four buttons one by one. | Each button click leads to its correct screen or action. |               | P   |
| <code>actionPerformed(ActionEvent)</code> | 1 | Create button opens Create screen | 1. Click the Create button.                   | Pirate Crew Create screen is shown.                      |               | P   |
|                                           | 2 | Create button updates info text   | 1. Click the Create button.                   | Info text changes to 'Pirate Crew Create'.               |               | P   |

|  |   |                                     |                                                        |                                                       |  |   |
|--|---|-------------------------------------|--------------------------------------------------------|-------------------------------------------------------|--|---|
|  | 3 | Create button refreshes pirate list | 1. Add a new Pirate.<br>2. Click the Create button.    | Create screen's captain list includes the new Pirate. |  | P |
|  | 4 | View button opens View screen       | 1. Click the View button.                              | Pirate Crew View screen is shown.                     |  | P |
|  | 5 | View button refreshes crew list     | 1. Add a new Pirate Crew.<br>2. Click the View button. | View screen's combo box includes the new crew.        |  | P |
|  | 6 | Edit button opens Modify screen     | 1. Click the Edit button.                              | Pirate Crew Modify screen is shown.                   |  | P |

|  |    |                                      |                                                                                                                    |                                                         |  |   |
|--|----|--------------------------------------|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|--|---|
|  | 7  | Edit button refreshes crew list      | <ol style="list-style-type: none"> <li>1. Add a new Pirate Crew.</li> <li>2. Click the Edit button.</li> </ol>     | Modify screen's combo box includes the new crew.        |  | P |
|  | 8  | Back button hides the menu           | <ol style="list-style-type: none"> <li>1. Click the Back button.</li> </ol>                                        | Pirate Crew panel becomes hidden.                       |  | P |
|  | 9  | Back button updates info text        | <ol style="list-style-type: none"> <li>1. Click the Back button.</li> </ol>                                        | Info text changes to 'Main Menu'.                       |  | P |
|  | 10 | Button sequence works with no errors | <ol style="list-style-type: none"> <li>1. Click Create.</li> <li>2. Click Back.</li> <li>3. Click Edit.</li> </ol> | Screens change correctly at each step, with no crashes. |  | P |

|  |    |                              |                                                      |                           |  |   |
|--|----|------------------------------|------------------------------------------------------|---------------------------|--|---|
|  | 11 | Unrelated click does nothing | 1. Trigger an action event from an unrelated source. | No screen change happens. |  | P |
|--|----|------------------------------|------------------------------------------------------|---------------------------|--|---|

### Class: PirateCrewCreateController

| Method        | # | Test Description                      | Sample Input Data                     | Expected Output                                    | Actual Output                                      | P/F |
|---------------|---|---------------------------------------|---------------------------------------|----------------------------------------------------|----------------------------------------------------|-----|
| refreshData() | 1 | Captain list is cleared before reload | 1. Call refreshData() twice in a row. | Captain combo box does not show duplicate entries. | Captain combo box does not show duplicate entries. | P   |
|               | 2 | 'None' is the first captain choice    | 1. Call refreshData().                | First item in captain combo box is 'None'.         | First item in captain combo box is 'None'.         | P   |

|                              |   |                                    |                                                                         |                                                                  |                                                                  |   |
|------------------------------|---|------------------------------------|-------------------------------------------------------------------------|------------------------------------------------------------------|------------------------------------------------------------------|---|
|                              | 3 | Only Pirates are listed            | 1. Add both Pirate and non-Pirate characters.<br>2. Call refreshData(). | Only Pirate characters appear in the captain list and checklist. | Only Pirate characters appear in the captain list and checklist. | P |
|                              | 4 | Checklist shows one row per Pirate | 1. Add three Pirates.<br>2. Call refreshData().                         | Checklist panel shows three selectable rows.                     | Checklist panel shows three selectable rows.                     | P |
| actionPerformed(ActionEvent) | 1 | Create with valid name and ship    | 1. Enter a crew name.<br>2. Enter a ship name.<br>3. Click Submit.      | New Pirate Crew is saved to the database.                        | New Pirate Crew is saved to the database.                        | P |
|                              | 2 | Create with a chosen captain       | 1. Pick a Pirate as captain.<br>2. Click Submit.                        | New crew's captain is set to that Pirate.                        | New crew's captain is set to that Pirate.                        | P |

|  |   |                               |                                                            |                                                                              |                                                                              |   |
|--|---|-------------------------------|------------------------------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------|---|
|  | 3 | Chosen captain role updates   | 1. Pick a Pirate as captain.<br>2. Click Submit.           | That Pirate's role becomes 'Captain' and its 'is captain' flag becomes true. | That Pirate's role becomes 'Captain' and its 'is captain' flag becomes true. | P |
|  | 4 | Create with 'None' as captain | 1. Leave captain as 'None'.<br>2. Click Submit.            | New crew has no captain set.                                                 | New crew has no captain set.                                                 | P |
|  | 5 | Checked members are added     | 1. Check two Pirates in the checklist.<br>2. Click Submit. | Both Pirates are added as crew members.                                      | Both Pirates are added as crew members.                                      | P |
|  | 6 | Successful create message     | 1. Fill the form correctly.<br>2. Click Submit.            | 'Pirate Crew Created Successfully!' message is shown.                        | 'Pirate Crew Created Successfully!' message is shown.                        | P |

|  |    |                                       |                                                      |                                                     |                                                     |   |
|--|----|---------------------------------------|------------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------|---|
|  | 7  | Screen closes after successful create | 1. Fill the form correctly.<br>2. Click Submit.      | Create screen closes and Pirate Crew menu is shown. | Create screen closes and Pirate Crew menu is shown. | P |
|  | 8  | Exit button cancels the form          | 1. Click the Exit button.                            | Create screen closes with no new crew saved.        | Create screen closes with no new crew saved.        | P |
|  | 9  | Error during create shows a message   | 1. Cause an error while submitting (ex: bad data).   | An error dialog shows the problem's message.        | An error dialog shows the problem's message.        | P |
|  | 10 | Database is saved after submit        | 1. Click Submit, whether it works or shows an error. | Character and Pirate Crew databases are both saved. | Character and Pirate Crew databases are both saved. | P |

|  |    |                             |                                                                         |                                                     |                                                     |   |
|--|----|-----------------------------|-------------------------------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------|---|
|  | 11 | Create with blank ship name | 1. Leave ship field blank.<br>2. Enter a crew name.<br>3. Click Submit. | New crew is still created, with an empty ship name. | New crew is still created, with an empty ship name. | P |
|--|----|-----------------------------|-------------------------------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------|---|

### Class: PirateCrewModifyController

| Method                       | # | Test Description                | Sample Input Data                  | Expected Output                                     | Actual Output                                       | P/F |
|------------------------------|---|---------------------------------|------------------------------------|-----------------------------------------------------|-----------------------------------------------------|-----|
| actionPerformed(ActionEvent) | 1 | Selecting a crew fills the form | 1. Pick a crew from the combo box. | Name and ship fields fill in with that crew's data. | Name and ship fields fill in with that crew's data. | P   |
|                              | 2 | Selecting a crew with a captain | 1. Pick a crew that has a captain. | Captain combo box shows the right captain selected. | Captain combo box shows the right captain selected. | P   |

|  |   |                                  |                                                                               |                                                             |                                                             |   |
|--|---|----------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------|---|
|  | 3 | Selecting a crew with no captain | 1. Pick a crew with no captain.                                               | Captain combo box shows 'None' selected.                    | Captain combo box shows 'None' selected.                    | P |
|  | 4 | Member checkboxes match the crew | 1. Pick a crew with known members.                                            | Checklist shows those members checked and others unchecked. | Checklist shows those members checked and others unchecked. | P |
|  | 5 | Update crew name                 | 1. Check the name checkbox.<br>2. Change the name text.<br>3. Click Submit.   | Crew's name is updated to the new text.                     | Crew's name is updated to the new text.                     | P |
|  | 6 | Name box left unchecked          | 1. Change the name text but leave the checkbox unchecked.<br>2. Click Submit. | Crew's name does not change.                                | Crew's name does not change.                                | P |

|  |    |                  |                                                                                                                                                              |                                              |                                              |   |
|--|----|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|----------------------------------------------|---|
|  | 7  | Update ship name | <ol style="list-style-type: none"> <li>1. Check the ship checkbox.</li> <li>2. Change the ship text.</li> <li>3. Click Submit.</li> </ol>                    | Crew's ship name is updated to the new text. | Crew's ship name is updated to the new text. | P |
|  | 8  | Update captain   | <ol style="list-style-type: none"> <li>1. Check the captain checkbox.</li> <li>2. Pick a new captain.</li> <li>3. Click Submit.</li> </ol>                   | Crew's captain changes to the new pick.      | Crew's captain changes to the new pick.      | P |
|  | 9  | Add a member     | <ol style="list-style-type: none"> <li>1. Check the members checkbox.</li> <li>2. Check a Pirate that was not a member.</li> <li>3. Click Submit.</li> </ol> | That Pirate is added to the crew.            | That Pirate is added to the crew.            | P |
|  | 10 | Remove a member  | <ol style="list-style-type: none"> <li>1. Check the members checkbox.</li> <li>2. Uncheck a Pirate that was a member.</li> <li>3. Click Submit.</li> </ol>   | That Pirate is removed from the crew.        | That Pirate is removed from the crew.        | P |

|                           |    |                                     |                                                    |                                                          |                                                          |   |
|---------------------------|----|-------------------------------------|----------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------|---|
|                           | 11 | Successful modify message           | 1. Make a change.<br>2. Click Submit.              | 'Pirate Crew Modified Successfully!' message is shown.   | 'Pirate Crew Modified Successfully!' message is shown.   | P |
|                           | 12 | Exit button cancels changes         | 1. Click the Exit button.                          | Modify screen closes and Pirate Crew menu is shown.      | Modify screen closes and Pirate Crew menu is shown.      | P |
|                           | 13 | Error during modify shows a message | 1. Cause an error while submitting.                | 'Modification Error' dialog shows the problem's message. | 'Modification Error' dialog shows the problem's message. | P |
| refreshComboBox()<br><br> | 1  | Refresh with no crews in database   | 1. Remove all crews.<br>2. Call refreshComboBox(). | All fields and the checklist are cleared.                | All fields and the checklist are cleared.                | P |

|  |   |                                 |                                                    |                                                    |                                                    |   |
|--|---|---------------------------------|----------------------------------------------------|----------------------------------------------------|----------------------------------------------------|---|
|  | 2 | Refresh reloads the Pirate list | 1. Add a new Pirate.<br>2. Call refreshComboBox(). | Captain list and checklist include the new Pirate. | Captain list and checklist include the new Pirate. | P |
|--|---|---------------------------------|----------------------------------------------------|----------------------------------------------------|----------------------------------------------------|---|

### Class: PirateCrewViewController

| Method                       | # | Test Description                          | Sample Input Data                                             | Expected Output                                                                | Actual Output                                                                  | P/F |
|------------------------------|---|-------------------------------------------|---------------------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-----|
| actionPerformed(ActionEvent) | 1 | Back button goes back to Pirate Crew menu | 1. Open Pirate Crew View screen.<br>2. Click the Back button. | Display changes to the Pirate Crew menu and the info text shows 'Pirate Crew'. | Display changes to the Pirate Crew menu and the info text shows 'Pirate Crew'. | P   |
|                              | 2 | Selecting a crew shows its name           | 1. Pick a crew from the combo box.                            | Name label shows 'Pirate Crew Name: <crew name>'.                              | Name label shows 'Pirate Crew Name: <crew name>'.                              | P   |

|  |   |                                      |                                         |                                                       |                                                       |   |
|--|---|--------------------------------------|-----------------------------------------|-------------------------------------------------------|-------------------------------------------------------|---|
|  | 3 | Selecting a crew shows its ship name | 1. Pick a crew from the combo box.      | Ship label shows 'Ship's Name: <ship name>'.          | Ship label shows 'Ship's Name: <ship name>'.          | P |
|  | 4 | Selecting a crew shows total bounty  | 1. Pick a crew from the combo box.      | Total bounty label shows the correct total amount.    | Total bounty label shows the correct total amount.    | P |
|  | 5 | Selecting a crew with a captain      | 1. Pick a crew that has a captain set.  | Captain label shows 'Captain's Name: <captain name>'. | Captain label shows 'Captain's Name: <captain name>'. | P |
|  | 6 | Selecting a crew with no captain     | 1. Pick a crew that has no captain set. | Captain label shows 'Captain's Name: None'.           | Captain label shows 'Captain's Name: None'.           | P |

|                       |   |                                 |                                                                   |                                                       |                                                       |   |
|-----------------------|---|---------------------------------|-------------------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------|---|
|                       | 7 | Member panels show up correctly | 1. Pick a crew that has members.                                  | Inner panel shows one panel for each crew member.     | Inner panel shows one panel for each crew member.     | P |
|                       | 8 | Switching between two crews     | 1. Select Crew A.<br>2. Select Crew B.                            | Labels update to show Crew B's details, not Crew A's. | Labels update to show Crew B's details, not Crew A's. | P |
|                       | 9 | Selecting the same crew again   | 1. Select a crew.<br>2. Select the same crew again.               | Labels stay correct and no error happens.             | Labels stay correct and no error happens.             | P |
| refreshComboBox()<br> | 1 | Combo box lists all crews       | 1. Add crews to the database.<br>2. Open Pirate Crew View screen. | Combo box shows each crew's ID and name.              | Combo box shows each crew's ID and name.              | P |

|  |   |                                                      |                                                                   |                                                                    |                                                                    |   |
|--|---|------------------------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------|---|
|  | 2 | Refresh combo box picks first crew                   | 1. Call refreshComboBox() while crews exist.                      | First crew in the list is auto-selected and its details are shown. | First crew in the list is auto-selected and its details are shown. | P |
|  | 3 | Refresh combo box with empty database                | 1. Make sure database has no crews.<br>2. Call refreshComboBox(). | All labels are cleared (name, ship, bounty, captain).              | All labels are cleared (name, ship, bounty, captain).              | P |
|  | 4 | Refresh combo box with empty database clears members | 1. Make sure database has no crews.<br>2. Call refreshComboBox(). | Inner panel has no member panels left.                             | Inner panel has no member panels left.                             | P |
|  | 5 | New crew shows up after refresh                      | 1. Add a new crew to the database.<br>2. Call refreshComboBox().  | New crew appears in the combo box list.                            | New crew appears in the combo box list.                            | P |

## Class: BountyController

| Method                                        | # | Test Description                                                                  | Sample Input Data                 | Expected Output                                                                                    | Actual Output                                                                                      | P/F |
|-----------------------------------------------|---|-----------------------------------------------------------------------------------|-----------------------------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-----|
| actionPerformed(ActionEvent) - registerButton | 1 | Clicking Register sets info text and switches to the register panel               | bountyView.registerButton clicked | mainview.setInfoText("Registering a Bounty") called<br>mainview.showPanel("BOUNTYREGISTER") called | mainview.setInfoText("Registering a Bounty") called<br>mainview.showPanel("BOUNTYREGISTER") called | P   |
|                                               | 2 | Clicking Register refreshes the target/captor combo boxes first                   | bountyView.registerButton clicked | bountyRegisterController.refreshComboBoxes() is invoked before showPanel() is called               | bountyRegisterController.refreshComboBoxes() is invoked before showPanel() is called               | P   |
|                                               | 3 | Clicking Register does not also refresh the view sub-controller's capture history | bountyView.registerButton clicked | bountyViewController.refreshView() is never called as part of this branch                          | bountyViewController.refreshView() is never called as part of this branch                          | P   |

|                                                                        |   |                                                                               |                               |                                                                                                                        |                                                                                                                        |   |
|------------------------------------------------------------------------|---|-------------------------------------------------------------------------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|---|
| actionPerformed(<br>ActionEvent) -<br>viewButton                       | 1 | Clicking View sets info text and switches to the view panel                   | bountyView.viewButton clicked | mainview.setInfoText("Viewing a Historical Bounty") called<br>mainview.showPanel("BOUNTYVIEW") called                  | mainview.setInfoText("Viewing a Historical Bounty") called<br>mainview.showPanel("BOUNTYVIEW") called                  | P |
|                                                                        | 2 | Clicking View refreshes the historical captures list first                    | bountyView.viewButton clicked | bountyViewController.refreshView() is invoked before showPanel() is called                                             | bountyViewController.refreshView() is invoked before showPanel() is called                                             | P |
|                                                                        | 3 | Clicking View does not also refresh the register sub-controller's combo boxes | bountyView.viewButton clicked | bountyRegisterController.refreshComboBoxes() is never called as part of this branch                                    | bountyRegisterController.refreshComboBoxes() is never called as part of this branch                                    | P |
| actionPerformed(<br>ActionEvent) -<br>backButton /<br>unmatched source | 1 | Clicking Back hides the Bounty panel and resets info text without navigating  | bountyView.backButton clicked | mainview.setInfoText("Main Menu") called<br>bountyView.panel.setVisible(false) no<br>mainview.showPanel() call is made | mainview.setInfoText("Main Menu") called<br>bountyView.panel.setVisible(false) no<br>mainview.showPanel() call is made | P |

|                                           |   |                                                                               |                                                               |                                                                                                                                               |                                                                                                                                               |   |
|-------------------------------------------|---|-------------------------------------------------------------------------------|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|---|
|                                           | 2 | Clicking Back does not refresh either sub-controller                          | bountyView.backButton clicked                                 | Neither refreshComboBoxes() nor refreshView() is called                                                                                       | Neither refreshComboBoxes() nor refreshView() is called                                                                                       | P |
|                                           | 3 | Clicking Back repeatedly is idempotent                                        | bountyView.backButton clicked twice in a row                  | The panel remains hidden and info text stays "Main Menu" after each click, with no exception                                                  | The panel remains hidden and info text stays "Main Menu" after each click, with no exception                                                  | P |
|                                           | 4 | An ActionEvent from a source not registered to this controller is ignored     | actionPerformed() called with an unrelated ActionEvent source | No info text change, no panel switch, and no exception is thrown                                                                              | No info text change, no panel switch, and no exception is thrown                                                                              | P |
| actionPerformed(ActionEvent) - sequencing | 1 | Clicking Register then View in sequence updates info text correctly each time | registerButton clicked, then viewButton clicked               | Info text reads "Registering a Bounty" after the first click and "Viewing a Historical Bounty" after the second, with no stale text left over | Info text reads "Registering a Bounty" after the first click and "Viewing a Historical Bounty" after the second, with no stale text left over | P |

|  |   |                                                                                                                          |                                                                              |                                                                                                                                            |                                                                                                                                            |   |
|--|---|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|---|
|  | 2 | Register/View branches and the Back branch are independent if-statements, so one ActionEvent triggers at most one branch | A single ActionEvent from viewButton                                         | Only the viewButton branch executes the unconditional backButton "if" check evaluates false and has no side effects                        | Only the viewButton branch executes the unconditional backButton "if" check evaluates false and has no side effects                        | P |
|  | 3 | Clicking Register when charDatabase has no eligible Pirates or Captors still shows the panel                             | charDatabase is empty<br>registerButton clicked                              | refreshComboBoxes() runs (producing empty combo boxes) and showPanel("BOUNTYREGISTER") is still called                                     | refreshComboBoxes() runs (producing empty combo boxes) and showPanel("BOUNTYREGISTER") is still called                                     | P |
|  | 4 | Rapid sequential clicks (Register -> Back -> View) leave state consistent with only the most recent click                | registerButton, then backButton, then viewButton clicked in quick succession | Final info text reads "Viewing a Historical Bounty" and the active panel is "BOUNTYVIEW", with no leftover effects from the earlier clicks | Final info text reads "Viewing a Historical Bounty" and the active panel is "BOUNTYVIEW", with no leftover effects from the earlier clicks | P |
|  | 5 | addActionListener() wires exactly the register, view, and back buttons and no other components                           | new BountyController(...) followed by inspecting registered listeners        | Only registerButton, viewButton, and backButton have this controller registered as an ActionListener                                       | Only registerButton, viewButton, and backButton have this controller registered as an ActionListener                                       | P |

## Class: BountyRegisterController

| Method                                                 | # | Test Description                                                      | Sample Input Data                                                         | Expected Output                                                                                              | Actual Output                                                                                                | P/F |
|--------------------------------------------------------|---|-----------------------------------------------------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-----|
| <code>actionPerformed(ActionEvent) - exitButton</code> | 1 | Clicking Exit sets info text and returns to the Bounty menu           | bountyRegisterView.exitButton clicked                                     | mainview.setInfoText("Bounty") called<br>mainview.showPanel("BOUNTY") called                                 | mainview.setInfoText("Bounty") called<br>mainview.showPanel("BOUNTY") called                                 | P   |
| <code>refreshComboBoxes()</code>                       | 1 | Refreshing populates the captured dropdown with only non-dead Pirates | charDatabase contains Pirates with status "Alive", "Captured", and "Dead" | capturedComboBox and availableTargets include only the Pirates whose status is not "Dead" (case-insensitive) | capturedComboBox and availableTargets include only the Pirates whose status is not "Dead" (case-insensitive) | P   |
|                                                        | 2 | Refreshing excludes dead Pirates entirely                             | A Pirate with <code>GetStatus() == "Dead"</code>                          | That Pirate does not appear in capturedComboBox or in the availableTargets list                              | That Pirate does not appear in capturedComboBox or in the availableTargets list                              | P   |

|                                             |   |                                                                                      |                                                                                      |                                                                                                                                                                     |                                                                                                                                                                     |   |
|---------------------------------------------|---|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|                                             | 3 | Refreshing populates the captor dropdown with Marines, Pirate Hunters, and Civilians | charDatabase contains a mix of Marine, PirateHunter, Civilian, and Pirate characters | captorComboBox and availableCaptors include every Marine/PirateHunter/Civilian Pirates are excluded from the captor list                                            | captorComboBox and availableCaptors include every Marine/PirateHunter/Civilian Pirates are excluded from the captor list                                            | P |
|                                             | 4 | Refreshing clears prior entries before repopulating                                  | refreshComboBoxes() called twice in a row with no data changes in between            | capturedComboBox and captorComboBox each contain exactly one entry per eligible character - no duplicates from the second call                                      | capturedComboBox and captorComboBox each contain exactly one entry per eligible character - no duplicates from the second call                                      | P |
| actionPerformed(ActionEvent) - submitButton | 1 | Submitting with both a target and captor selected registers the capture              | capturedComboBox and captorComboBox both have a valid ( $\geq 0$ ) selection         | bountyDatabase.registerCapture(captor, target, "captured") is called with the Pirate and Character resolved from availableTargets/availableCaptors at those indices | bountyDatabase.registerCapture(captor, target, "captured") is called with the Pirate and Character resolved from availableTargets/availableCaptors at those indices | P |
|                                             | 2 | A successful registration shows a confirmation naming the captor                     | Valid selections, captor named "Smoker"                                              | JOptionPane message reads "Bounty successfully claimed by Smoker!"                                                                                                  | JOptionPane message reads "Bounty successfully claimed by Smoker!"                                                                                                  | P |

|  |   |                                                                        |                                                                                     |                                                                                                                             |                                                                                                                             |   |
|--|---|------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---|
|  | 3 | A successful registration returns to the Bounty menu                   | Valid selections                                                                    | mainview.setInfoText("Bounty") called<br>mainview.showPanel("BOUNTY") called                                                | mainview.setInfoText("Bounty") called<br>mainview.showPanel("BOUNTY") called                                                | P |
|  | 4 | Submitting with no target selected is rejected                         | capturedComboBox selection = -1,<br>captorComboBox selection = valid                | A generic Exception is thrown and caught error dialog reads "Error: Please ensure both a Target and a Captor are selected." | A generic Exception is thrown and caught error dialog reads "Error: Please ensure both a Target and a Captor are selected." | P |
|  | 5 | Submitting with no captor selected is rejected                         | captorComboBox selection = -1,<br>capturedComboBox selection = valid                | Same validation exception is thrown and caught identical error dialog is shown                                              | Same validation exception is thrown and caught identical error dialog is shown                                              | P |
|  | 6 | An exception thrown by registerCapture() itself is caught and reported | bountyDatabase.registerCapture() throws an exception (ex: duplicate capture record) | Caught by the same generic Exception handler dialog reads "Error: " + that exception's message, titled "Registration Error" | Caught by the same generic Exception handler dialog reads "Error: " + that exception's message, titled "Registration Error" | P |

|  |   |                                                                                                                            |                                                                                                                         |                                                                                                                                                                             |                                                                                                                                                                             |   |
|--|---|----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|  | 7 | Submitting before refreshComboBoxes() has ever populated<br>availableTargets/availableCaptors is still handled gracefully  | submitButton clicked while<br>availableTargets/availableCaptors are null                                                | The resulting NullPointerException is still caught by the generic catch(Exception) block (NPE is a subclass of Exception) and shown as an error dialog rather than crashing | The resulting NullPointerException is still caught by the generic catch(Exception) block (NPE is a subclass of Exception) and shown as an error dialog rather than crashing | P |
|  | 8 | The bounty database is saved on every submit click, regardless of outcome                                                  | Submit with a valid registration, a missing selection, and a thrown registerCapture() exception (three separate clicks) | bountyDatabase.writeDatabase() is called once per click in all three cases, from the finally block                                                                          | bountyDatabase.writeDatabase() is called once per click in all three cases, from the finally block                                                                          | P |
|  | 9 | A Pirate whose status changes to "Dead" after the last refreshComboBoxes() call is still selectable until the next refresh | Call refreshComboBoxes(), change a listed Pirate's status to "Dead", then submit while still selecting that Pirate      | The stale in-memory availableTargets list still contains that Pirate, so the capture is registered using data as of the last refresh, not the character's current status    | The stale in-memory availableTargets list still contains that Pirate, so the capture is registered using data as of the last refresh, not the character's current status    | P |

|  |    |                                                                                                          |                                                                    |                                                                                                                |                                                                                                                |   |
|--|----|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|---|
|  | 10 | Submitting twice in a row for two different valid target/captor pairs registers two independent captures | Submit Target A/Captor A, then reopen and submit Target B/Captor B | bountyDatabase.registerCapture() is called once per submission with the correct, independent pairing each time | bountyDatabase.registerCapture() is called once per submission with the correct, independent pairing each time | P |
|--|----|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|---|

### Class: BountyViewController

| Method                                    | # | Test Description                                            | Sample Input Data                                            | Expected Output                                                              | Actual Output                                                                | P/F |
|-------------------------------------------|---|-------------------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------|-----|
| actionPerformed(ActionEvent) - backButton | 1 | Clicking Back sets info text and returns to the Bounty menu | bountyViewView.backButton clicked                            | mainview.setInfoText("Bounty") called<br>mainview.showPanel("BOUNTY") called | mainview.setInfoText("Bounty") called<br>mainview.showPanel("BOUNTY") called | P   |
|                                           | 2 | An ActionEvent from an unrelated source is ignored          | actionPerformed() called with a source other than backButton | No info text change, no panel switch, and no exception is thrown             | No info text change, no panel switch, and no exception is thrown             | P   |

|               |   |                                                                                                                                     |                                                                                     |                                                                                                                            |                                                                                                                            |   |
|---------------|---|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|---|
|               | 3 | Clicking Back does not call refreshView()                                                                                           | bountyViewView.backButton clicked                                                   | bountyViewView.setHistoricalCaptures() is never invoked as a side effect of navigating back                                | bountyViewView.setHistoricalCaptures() is never invoked as a side effect of navigating back                                | P |
|               | 4 | Clicking Back repeatedly behaves the same each time                                                                                 | backButton clicked twice in a row                                                   | Info text is set to "Bounty" and showPanel("BOUNTY") is called on each click, with no exception or duplicated side effects | Info text is set to "Bounty" and showPanel("BOUNTY") is called on each click, with no exception or duplicated side effects | P |
|               | 5 | Clicking Back after refreshView() has already populated the panel still navigates away without altering the underlying capture data | refreshView() called, then backButton clicked                                       | Navigation occurs normally bountyDatabase.getCaptureHistory() is not re-queried or mutated by the Back action              | Navigation occurs normally bountyDatabase.getCaptureHistory() is not re-queried or mutated by the Back action              | P |
| refreshView() | 1 | Refreshing passes the current capture history to the view                                                                           | bountyDatabase.getCaptureHistory() returns a list of N records refreshView() called | bountyViewView.setHistoricalCaptures() is called with exactly that list                                                    | bountyViewView.setHistoricalCaptures() is called with exactly that list                                                    | P |

|  |   |                                                                 |                                                                |                                                                                                         |                                                                                                         |   |
|--|---|-----------------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|---|
|  | 2 | Refreshing triggers a UI revalidate after updating content      | refreshView() called                                           | bountyViewView.innerPanel.revalidate() is called immediately after setHistoricalCaptures()              | bountyViewView.innerPanel.revalidate() is called immediately after setHistoricalCaptures()              | P |
|  | 3 | Refreshing triggers a UI repaint after revalidating             | refreshView() called                                           | bountyViewView.innerPanel.repaint() is called immediately after revalidate()                            | bountyViewView.innerPanel.repaint() is called immediately after revalidate()                            | P |
|  | 4 | Refreshing with an empty capture history does not error         | bountyDatabase.getCaptureHistory() returns an empty list       | setHistoricalCaptures() is called with an empty list revalidate()/repaint() still run without exception | setHistoricalCaptures() is called with an empty list revalidate()/repaint() still run without exception | P |
|  | 5 | Refreshing does not change mainview's info text or active panel | refreshView() called directly, independent of any button click | mainview.setInfoText() and mainview.showPanel() are never called by refreshView() itself                | mainview.setInfoText() and mainview.showPanel() are never called by refreshView() itself                | P |

|  |   |                                                                                                                          |                                                                                                                                   |                                                                                                                                                              |                                                                                                                                                              |   |
|--|---|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|  | 6 | Calling <code>refreshView()</code> twice in a row reflects any new captures recorded between calls                       | Call <code>refreshView()</code> , register a new capture via the Bounty Register flow, then call <code>refreshView()</code> again | The second call's <code>setHistoricalCaptures()</code> list includes the newly added capture record - the data is re-fetched, not cached from the first call | The second call's <code>setHistoricalCaptures()</code> list includes the newly added capture record - the data is re-fetched, not cached from the first call | P |
|  | 7 | The list passed to <code>setHistoricalCaptures()</code> preserves the order returned by <code>getCaptureHistory()</code> | <code>bountyDatabase.getCaptureHistory()</code> returns records in chronological order                                            | <code>setHistoricalCaptures()</code> receives that same list reference/order, unmodified                                                                     | <code>setHistoricalCaptures()</code> receives that same list reference/order, unmodified                                                                     | P |
|  | 8 | <code>refreshView()</code> can be invoked independent of any button click and still updates the panel                    | <code>refreshView()</code> called directly without any prior <code>ActionEvent</code>                                             | The panel's historical captures display updates correctly with no dependency on <code>actionPerformed()</code> having run first                              | The panel's historical captures display updates correctly with no dependency on <code>actionPerformed()</code> having run first                              | P |
|  | 9 | <code>addActionListener()</code> wires exactly the back button and no other components                                   | <code>new BountyViewController(...)</code> followed by inspecting registered listeners                                            | Only <code>backButton</code> has this controller registered as an <code>ActionListener</code>                                                                | Only <code>backButton</code> has this controller registered as an <code>ActionListener</code>                                                                | P |

|  |    |                                                                                                   |                                                                                 |                                                                        |                                                                        |   |
|--|----|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------|---|
|  | 10 | refreshView() with a very large capture history still passes the complete list without truncation | bountyDatabase.getCaptureHistory() returns 100+ records<br>refreshView() called | setHistoricalCaptures() receives the full list with no records dropped | setHistoricalCaptures() receives the full list with no records dropped | P |
|--|----|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------|---|